

Research papers

Optimal scheduling of hydropower and pumped storage hydropower for high renewable energy share in Thailand: A novel hybrid optimization approach with dimensionality reduction

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ARTICLE INFO

Keywords:

Pumped storage hydropower
Renewable energy
Compressed sensing
Optimization
Scheduling operation

ABSTRACT

The large-scale integration of Renewable Energy (RE) into national power grid presents complex challenges in balancing intermittency, system reliability, and economic viability. Conventional optimization methods often do not cover the nonlinear, high-dimensional, and multi-objective of RE integration with Energy Storage System (ESS) scheduling operation under real-world constraints. This study explores the optimal development and operation of RE systems—including solar, wind, small hydropower, and 12 conventional hydropower integrated with Pumped Storage Hydropower (PSH) as an energy storage solution—operated by the Electricity Generating Authority of Thailand (EGAT). To address these challenges, A hybrid metaheuristic optimization framework, Hybrid Genetic Algorithm (GA) – Particle Swarm Optimization (PSO) with Compressed Sensing (CS) Dimensionality Reduction, is developed to enhance the planning and scheduling of power dispatch from solar PV, wind power, and PSH systems. The model aims to maximize RE share and minimize Levelized Cost of Electricity (LCOE), while satisfying annual electricity demand and system reliability constraints. Importantly, this research incorporates reservoir simulation and water resources management, considering real-world operational constraints of EGAT's hydropower reservoirs. By doing so, the optimized scheduling operation ensures higher practical feasibility and alignment with actual hydrological limitations. These results confirmed that a 37.87% of RE share can be achieved without curtailment by optimally coordinating hydropower, and PSH to align with solar PV and wind power generation. However, the findings revealed a critical trade-off between increasing RE share and system-wide LCOE: as RE share rises from 8.70% to 37.87%, LCOE increases from 1.37 THB/kWh to 1.89 THB/kWh due to overcapacity of solar PV and wind installed capacity. While solar PV and wind remain low-cost individually, system integration at high RE share leads to diminishing economic returns. Therefore, the proposed hybrid scheduling model not only supports higher RE integration but also provides a cost-effectiveness benchmark for future power system planning in Thailand.

Nomenclatures

Abbreviations and acronyms

BESS	Battery Energy Storage System
CS	Compressed Sensing
EGAT	Electricity Generating Authority of Thailand
ERC	Energy Regulatory Commission
ESS	Energy Storage System
FPV	Floating Photovoltaic
GA	Genetic Algorithm

LOLE	Loss of Load Expectation
NCC	National Control Center
NPV	Net Present Value
NSGA-II	Non-dominated Sorting Genetic Algorithm II
PCA	Principal Component Analysis
PSH	Pumped Storage Hydropower
PSO	Particle Swarm Optimization
PV	Photovoltaic
RE	Renewable Energy
RID	Royal Irrigation Department
RIP	Restricted Isometry Property

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<https://doi.org/10.1016/j.est.2026.120842>

Received 5 September 2025; Received in revised form 14 January 2026; Accepted 27 January 2026

Available online 6 February 2026

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ROIC Return on Invested Capital
VRE Variable Renewable Energy

Symbols

a Incidence-weighted solid angle as seen by the tilted surface [–]
 A Surface area [m²]
 $Albedo$ Diffuse reflection of solar radiation from the ground [–]
 AOI Angle of incidence of beam radiation on a surface [°]
 b Incidence-weighted solid angle as seen by the horizontal surface [–]
 B_n Nth benefit function [–]
 c_1 Learning factors according to the best position [–]
 c_2 Learning factors according to position of the best particle [–]
 $C_{Capital}$ Annualized initial capital costs [THB/kW]
 C_{Fuel} Annualized fuel costs [THB/kW]
 $C_{O\&M}$ Annualized Operation and Maintenance (O&M) costs [THB/kW]
 C_p Power coefficient of the wind turbine [–]
 D Spillway discharge [m³/h]
 DHI Diffuse horizontal irradiance [W/m²]
 DNI Direct normal irradiance [W/m²]
 E_{Served} Total annualized energy generation [kWh]
 FC Fuel cost of thermal power plants [THB/kW]
 F_1 Perez model circumsolar anisotropy coefficient [–]
 F_2 Perez model horizontal/zenith anisotropy coefficient [–]
 g Gravitational acceleration [m/s²]
 g^{best} Position of the best particle [–]
 H Net head of the reservoir [m.]
 HR Net plant heat rate [BTU/kWh]
 i Discount rate [%]
 I Natural flow from river [m³/h]
 I_{gr} Ground reflect irradiance [W/m²]
 I_{diff} Sky diffuse irradiance [W/m²]
 I_{dir} Beam irradiance [W/m²]
 IC Installed capacity [MW]
 $LCOE$ Levelized Cost of Electricity [THB/kWh]
 N Number of units of power plants [–]
 OM Operation & Maintenance [THB/kW]
 P Power [MW]
 POA Total irradiance from the solar energy [W/m²]
 POA_{STC} Total irradiance from the solar energy at Standard Test Conditions [W/m²]
 PR Performance ratio of solar PV array [–]
 p_z^{best} Best position in compressed domain [–]
 Q_{in} Volumetric flow rate for pumping water [m³/h]
 Q_{out} Volumetric flow rate for discharge generating water [m³/h]
 r Random variables in PSO [–]
 TAC Total annualized cost [THB]
 v Wind velocity [m/s]
 $v_{z,i}$ Velocity in compressed domain of each particle [m/s]
 v_{Cut-in} Cut-in wind velocity [m/s]
 $v_{Cut-out}$ Cut-out wind velocity [m/s]
 v_{Rated} Wind turbine rated velocity [m/s]
 V Storage volume [m³]
 w_{B_n} Weight for number of benefit function B_n [–]
 w_{PSO} Learning factor according to the velocity of the current state in PSO [–]
 $\overline{WR}$ Annual average water rate [m³/kWh]
 x_i Original high-dimensional decision vector [–]
 x^* Reconstructed decision vector [–]
 y Measurement vector in compressed domain [–]

y^* Optimal solution from Hybrid GA-PSO [–]
 z_i Position vector of each particle i at the current iteration t in compressed domain [–]
 $z_{offspring}$ Decision variables of offspring in compressed domain

Greeks

α Specific investment cost [THB/kW]
 β Slope of tilted angle [°]
 Δt Time duration of time step [h]
 γ Surface azimuth angle [°]
 η Efficiency [%]
 θ_z Zenith angle [°]
 θ Sparse coefficient vector [–]
 θ^* Solution of the sparse recovery problem [–]
 μ Mutation rate [%]
 ρ Density of the air [kg/m³]
 Φ Sensing Matrix [–]
 Ψ Basis matrix [–]

Superscripts

K Sparse coefficients in compressed sensing [–]
 n_{ix} Dimension of original decision vector [–]
 n_{iter} Number of iterations in the optimization process [–]

Subscripts

$actual$ Actual data [–]
 gen Generating mode [–]
 GSA Global Solar Atlas [–]
 h Hydropower plants [–]
 l Load demand [–]
 $pump$ Pumping mode [–]
 psh Pumped Storage Hydropower [–]
 s Solar PV plants [–]
 sh Small hydropower plants [–]
 sim Simulated data [–]
 th Thermal power plants [–]
 w Wind power plants [–]
 x Power plant types [–]

1. Introduction

A typical strategy for increasing Renewable Energy (RE) is the large-scale deployment of solar and wind energy; however, this approach faces several technical challenges. First, their variability and weather-dependent, leading to power intermittency and difficulties in balancing supply and demand in real time [1]. Second, geographical constraints may not align with demand centers, causing transmission congestion [2]. Third, without sufficient storage systems, high shares of these resources can also reduce system reliability [3]. The environmental impacts motivating RE adoption increases from the reliance on fossil fuels in global energy consumption—over 85%—which significantly contributes to CO₂ emissions and climate change. Thus, replacing fossil fuels with RE sources, particularly solar, wind, and hydropower, is essential to meet both environmental and economic goals. To address these issues, various approaches have been proposed, such as flexible power generation [4], diversification of RE generation [5], Cross-Border interconnection [6], Curtail excess generation and Power-to-X [7], RE Integration [7], Smart grids [4], and Energy Storage Systems [8].

Many studies [9–14] evaluated the integration of RE sources with Energy Storage Systems (ESSs) to address the variability and intermittency of solar and wind power. ESSs, grid interconnection, and multi-source complementarity were shown to enhance power system

flexibility and reliability. Similarly, **Farah and Andresen [15]** demonstrated that the complementary use of multiple RE sources could effectively mitigate the fluctuations inherent in single-source systems. Moreover, integrated configurations such as solar–wind–PSH hybrids were shown to improve generation performance and reduce the Levelized Cost of Electricity (LCOE) compared to stand-alone or conventional systems. However, **Samende et al. [16]** cautioned that fully replacing conventional power plants with variable RE sources might introduce system stability issues, including frequency oscillations under disturbances. In a long-term perspective, fossil fuels are expected to be substituted by RE technologies, despite arguments that conventional power plants remain vital enablers of RE integration. **Javadi et al. [17]** investigated the optimal allocation of spinning reserve between thermal units and electrical energy storage systems (ESSs) in a 6-bus test system using a two-stage stochastic security-constrained unit commitment (SCUC) model. Their results demonstrated that incorporating ESSs significantly reduces operating costs by decreasing reliance on thermal units, while the fast-response capability of ESSs makes them effective reserve providers. Most research focuses on optimizing the scheduling operation of hybrid configurations to meet demand reliably and cost-effectively under real-world constraints.

Pumped Storage Hydropower (PSH) is the most mature and widely deployed ESS for large-scale RE integration, offering roundtrip efficiency of 70%–85% [18]. PSH is characterized by long lifetime 50–80 years [19] and negligible performance degradation over decades of operation, making it particularly suitable for long-term grid-scale energy balancing [20]. **Luo et al. [21]** reported that PSH units also enable energy shifting required for renewable integration. **Ramos et al. [22]** emphasized that PSH enables dispatchable power by storing excess energy and releasing it during peak demand, thereby supporting load balancing, frequency regulation, fast response, and curtailment reduction. PSH also demonstrates comparative advantages over other ESSs in terms of environmental and economic sustainability. **Singhal et al. [23]** stated that although Battery Energy Storage System (BESS) adoption is increasing, its environmental concerns related to toxic materials, making PSH a more environmentally and economically sustainable option, particularly in high-RE systems. Nonetheless, **Ba-abbad et al. [24]** reported that PSH deployment is constrained by high capital costs, long construction timelines, site-specific topography, potential environmental impacts, and limited capacity expansion of installed capacity during operation. Despite these challenges, **Naval et al. [25]** underscored that PSH remains the most cost-effective and reliable grid-scale storage technology for managing the variability of solar and wind power.

So far, there are many research investigating the ancillary services of PSH units in power systems, which plays a critical role in short-term scheduling operation. **Jordehi et al. [26]** highlighted that the availability PSH units in power system is primarily reflected in its operational and economic impact on microgrid operation. Their results showed that integrating PSH reduced the microgrid operating cost by approximately 3.65%, while enhancing energy arbitrage capability by charging during low-price periods and discharging during high-price periods, thereby reducing reliance on expensive thermal power generation. **Simab et al. [27]** proposed an efficient multi-objective framework for short-term hydrothermal scheduling operation including PSH units using Normal Boundary Intersection (NBI) method. The study found that PSH significantly improves hydro–thermal system performance, achieving multi-percent cost and emission reductions by enabling energy shifting and coordinated multi-objective scheduling. Another operational benefit of PSH units, reference from **Neshad et al. [28]**, is fast response capability enabled by rapid start-up and shut-down as well as wide operating flexibility, which allows participation in frequency control. However, its operation is constrained by the finite availability of water resources.

Recent studies [21,29–36] on hybrid RE systems integrating hydropower and PSH highlighted the importance of incorporating both physical and economic constraints and objective function in

optimization models. **Luo et al. [21]** applied technical constraints typically include water balance, reservoir limits, head variation, generation and pumping capacities, and environmental flow requirements, while economic constraints such as LCOE and Net Present Value (NPV) were used to ensure economic feasibility. However, most existing models were limited by simplifying assumptions (e.g., single-plant focus, lack of seasonal variation, or national-scale coordination) and linear optimization techniques that fail to capture nonlinear dynamics. To address these limitations, **Zhou et al. [29]** advocated the adoption of multi-objective optimization methods, particularly Genetic Algorithms (GA), to enhance system reliability (e.g., Loss of Load Expectation, LOLE) while simultaneously optimizing economic performance under real-world constraints. Similarly, **Tang et al. [30]** emphasized the need for multi-reservoir, high-resolution models that can capture the variability and operational interdependence of water resources in RE-integrated systems. Furthermore, **Nassar et al. [31]** underscored the critical role of hydrological constraints—such as water balance, flow rate, and reservoir levels—in shaping hydropower scheduling operation, and suggested that wind energy could effectively complement PSH operations during off-seasons or periods when hydropower faces irrigation or flood control restrictions. **Irshad et al. [32]** emphasized system reliability improvements and LCOE reduction on large-scale PSH primarily. However, **Wang et al. [33]** noted that limited studies have explored optimized scheduling operations for achieving high RE share in coordination with conventional power plants. To address emerging challenges, **Liu et al. [34]** proposed the integration of Floating Photovoltaic (FPV) and PSH systems to mitigate reliability issues arising from excess FPV generation and load imbalance. Building on this, **Xu et al. [35]** developed mathematical models to evaluate both the power system reliability and economic performance of variable RE sources integrated with PSH. Similarly, **Papaeftymiou et al. [36]** demonstrated the critical role of PSH in increasing RE share, particularly within small or isolated grids. Furthermore, **Xu et al. [35]** highlighted the importance of optimal scheduling in solar–wind–PSH hybrid systems for maintaining full operational capacity while balancing cost and reliability.

Most previous studies focused on regional areas, and the existing limited national-scale rarely consider an hourly power-balance that met actual load demand, implicitly allowing RE curtailment resulting in overestimating achievable RE shares. Furthermore, many scheduling studies ignored the coordinated operation of multiple-reservoir hydropower plants, and several works evaluated Loss of Load Probability (LPS)—LCOE trade-offs rather than identifying the maximum RE share. In addition, solar and wind capacities were often evaluated from resource potential instead of being optimally determined. These methods lead to them not being consistent with how to achieve the highest RE share. These limitations are critical in capacity-expansion studies, where optimal sizing can either unaccounted RE curtailment or underestimate the achievable RE share. Furthermore, most of previous studies did not address the high-dimensionality hourly scheduling operation, resulting in computationally intractable. These are our challenging problems investigated in this research.

In addition to conventional single optimization approaches, recent studies [30,37,38] have increasingly explored hybrid optimization methods for improving the scheduling operation of RE systems. **Tang et al. [30]** utilized Non-dominated Sorting Genetic Algorithm II (NSGA-II) for multi-objective optimization in PSH–BESS systems, and demonstrated its effectiveness in balancing trade-offs between competing objectives. Similarly, **Gan et al. [37]** reported high performance of hybrid Particle Swarm Optimization (PSO)–Cuckoo Search models in Solar–Wind–PSH applications. Although Multi-Objective Particle Swarm Optimization (MOPSO) has been widely adopted, its limited local search capability and slow convergence in fine-tuning stages constrain its applicability in large-scale problems. In contrast, **Toro et al. [38]** highlighted the capability of Genetic Algorithms (GA) for solving complex multi-objective and nonlinear problems. To improve the complementary strengths of both methods, Hybrid GA–PSO has emerged as a

promising approach that combines the global search ability and population diversity of GA with the fast convergence and local search of PSO. Therefore, this study employed a Hybrid GA-PSO Optimization to optimize the long-term scheduling of a Solar-Wind-PSH hybrid system, enabling better trade-off solutions across LCOE, and RE share while accommodating operational constraints.

This study aims to develop an hourly-resolution model for optimal scheduling operation a hybrid RE system—comprising solar, wind, small hydropower, and conventional hydropower integrated with PSH—alongside conventional power plants in Thailand. The objective is to design a cost-effective and reliable scheduling strategy through coordinated dispatch between renewable generation and energy storage, using mathematical modeling to ensure system adequacy. The primary goal is to maximize RE share by optimizing the installed capacities of solar PV and wind with the optimization of hydropower and PSH dispatch strategies across existing 12 major EGAT's hydropower and PSH. The coordinated model is developed to optimize hydropower and PSH dispatch to ensure efficient use of hydropower resources, thereby enabling the highest RE share. This research addresses critical knowledge gaps by introducing the first national-scale model that coordinates the operation of 12 EGAT hydropower and PSH plants under real-world operational reservoir constraints. To the best authors' knowledge, Compressed Sensing has not yet been applied with optimization frameworks for RE systems and PSH scheduling operation, and this study represents the first attempt to integrate Compressed Sensing (CS) into A hybrid GA-PSO. A hybrid GA-PSO with CS framework is employed to jointly optimize, using RE Share and LCOE as main objective functions. The model simulates a full-year hourly dispatch horizon to capture seasonal and diurnal variability and evaluates three scenarios to identify the most cost-effective generation portfolio for achieving the highest possible RE share. The paper's most important main contributions in this regard are as follows:

- Developing an integrated national-scale capacity expansion and coordinated multiple hydropower – PSH operation model, which each plant can be defined storage volume limit constraints separately.
- Presenting an optimization framework with dimensionality reduction for addressing multi-dimensionality problems of the hourly-based scheduling operation.
- Applying actual hourly load demand, and hydrological data to the multi-objective optimization framework to evaluate the model's coordinated power balance dispatch for maximizing RE share and minimizing LCOE.
- Using hourly historical 13 years (2012–2024) of actual operating data to validate the coordinated hydropower and PSH dispatch model.

The remainder of this paper is organized as follows. **Section 2** introduces the mathematical modeling framework and model validation. **Section 3** describes the scenario design and presents a case study based in Thailand. **Section 4** discusses the optimized dispatch strategies for a hybrid RE system, focusing on the coordinated operation of 12 major EGAT hydropower plants under scenario-based analysis, also analyzes the optimum installed capacities of solar and wind power for the case studies, and sensitivity analysis for LCOE. **Section 5** outlines the limitations of this study, and **Section 6** provides concluding remarks and directions for future research.

2. Methods for optimizing scheduling operation of the solar-wind-PSH hybrid system

The scope of this study is outlined in **Section 2.1**. **Section 2.2** presents the Coordinated Operation Model, the development of mathematical models for the hybrid RE system and the formulation used to calculate the LCOE is introduced. In **Section 2.3**, Hydropower Model Comparison is presented. Finally, **Section 2.4** the implementation of a

Hybrid Genetic Algorithm (GA) – Particle Swarm Optimization (PSO) with Compressed Sensing (CS) Dimensionality Reduction approach is proposed in the details.

2.1. Scope of this study

This study developed a long-term stochastic optimization model for

Table 1
List of current EGAT's Renewable Energy Power Plants.

Item	Project name	Type	Total installed capacity (MW)	Storage Capacity (MCM)	Remark
1	Lamtakong	Pure PSH	1000	10.33	Gen./pump
2	Jolabha Vadhana	Reservoir type with PSH	720	17745	360 MW (Gen. only)
3	Srinagarind Dam	Reservoir type with PSH	779.20	13462	360 MW (Gen./pump)
4	Bhumibol Dam	Reservoir type with PSH	779.20	13462	608.2 MW (Gen. only)
5	Rajjaprabha Dam	Reservoir type	240	5638.84	171 MW (Gen./pump)
6	Vajiralongkorn Dam	Reservoir type	300	8860	Gen. only
7	Bang Lang Dam	Reservoir type	84	1454.36	
8	Sirikit Dam	Reservoir type	500	9510	
9	Ubol Ratana Dam	Reservoir type	25.20	2431.30	
10	Sirindhorn Dam	Reservoir type	36	1966.47	
11	Chulabhorn Dam	Reservoir type	40	163.75	
12	Nam Pung Dam	Reservoir type	6	165.48	
13	Huai Kum Dam	Reservoir type	1.2	20.23	
14	Chao Phraya Dam	Run-off-River	6	–	–
15	Naresuan Dam	Run-off-River	8	–	–
16	Mae Klong Dam	Run-off-River	12	–	–
17	Khun Dan Prakan Chon Dam	Run-off-River	10	–	–
18	Pasak Jolasid Dam	Run-off-River	6.70	–	–
19	Kwae Noi Bumrung Dam	Run-off-River	30	–	–
20	Kiew Khor Mah Dam	Run-off-River	5.5	–	–
21	Khlong Tron Dam	Run-off-River	2.50	–	–
22	Chulabhorn Dam	Run-off-River	1.25	–	–
23	Pha Chuk Dam	Run-off-River	14	–	–
24	Sirindhorn Dam	Floating Solar	45	–	–
25	Ubol Ratana Dam	Floating Solar	24	–	–
26	Lam Ta Khong Wind Turbine	Wind	24	–	–

scheduling a hybrid energy system comprising solar, wind, and PSH to assess whether Thailand's existing and planned capacities (See Table 1). The model operated at hourly resolution over a full-year horizon, capturing temporal variability in both generation and demand. A generation-load balance framework was used to evaluate system adequacy, incorporating actual hourly hydrological data covering 2012–2024 provided by Electricity Generating Authority of Thailand (EGAT) Water Intelligence Center [39]. In this study, thermal power plants were scheduled to supply the residual load demand, thereby ensuring system adequacy during periods when Variable Renewable Energy (VRE) was insufficient to fully meet the load demand. The two main objectives of optimization were to maximize RE share and minimize LCOE. The model was applied to Thailand's national grid using data from 12's major EGAT hydropower, PSH plants, Solar, and Wind to evaluate their coordinated contribution to system reliability. Reservoir hydropower including PSH, downstream irrigation hydropower, and floating photovoltaic plants have total installed capacities of 3731.6 MW, 95.95 MW, and 69 MW, respectively, as summarized in Table 2. Additionally, historical electricity generation statistics from 2012 to 2024, recorded at 30-min intervals by EGAT [40], provided insights into national power generation trends across all fuel types. The resulting hourly datasets were then averaged across the 2012–2024 period to reflect historical load demand trends and account for long-term variability.

2.2. Mathematical model for solar-wind-PSH hybrid system

The scheduling operations of Solar PV, Wind, and Pumped Storage Hydropower (PSH) systems are typically developed independently, reflecting the inherent differences in their operational characteristics and constraints. To achieve optimal integration, a collaborative dispatching strategy adopted to coordinate the operation of the Solar–Wind–PSH hybrid system. Under this strategy, the aggregated output from solar and wind resources were prioritized to meet the electricity demand, followed by hydropower and thermal power to ensure overall power balance as:

$$P_s(t) + P_w(t) + P_h(t) + P_{sh}(t) + P_{th}(t) = P_l(t) + P_{psh}(t) \quad (1)$$

This equation explicitly represents the load demand balancing

mechanism for the hybrid Solar–Wind–PSH–Thermal system. It ensures that the total electricity demand, $P_l(t)$, at time t is satisfied by the combined contributions from solar power generation, $P_s(t)$, wind power generation, $P_w(t)$, hydropower generation, $P_h(t)$, small hydropower generation, $P_{sh}(t)$, and PSH pumping power generation, $P_{psh}(t)$. Any residual demand that remains unmet by renewable and hydropower sources is then fulfilled by thermal power generation $P_{th}(t)$, thereby guaranteeing supply-demand balance in all operating hours.

The hybrid Solar-Wind-PSH system is shown in Fig. 1. Hydropower is assigned as the second priority and is dispatched to meet the remaining demand after fully utilizing solar, wind and small hydropower resources. Its operation is subject to reservoir management constraints and water allocation policies to ensure sustainable usage over the scheduling horizon. Finally, thermal power generation is employed as a last resort to supply any residual demand that cannot be met by renewable and hydropower sources. Thermal power plants operate only when the combined renewable and hydropower generation is insufficient to balance the electricity demand, ensuring grid stability and avoiding load shedding. When renewable and hydropower generation are adequate, thermal generation remains offline, thereby minimizing LCOE.

The mathematical models for evaluating solar power, wind power, hydropower, and PSH power were constructed. The coordinated operation necessitates the development of distinct energy models. The energy models for solar, wind, and hydropower were developed to represent RE systems considered in this study. While the scheduling operations of Solar PV, Wind, and PSH are typically developed independently to reflect their distinct operational characteristics and constraints, their coordinated integration is critical for ensuring system-wide optimization.

2.2.1. Mathematical model for evaluating the solar PV power

The solar PV power output is determined as a function of solar irradiance. Following Sapthanakorn et al. [41] and Perez et al. [42], the total solar power output in this study is calculated from beam irradiance, sky diffuse irradiance, and ground reflect irradiance as described in Eq. 1– Eq.11.

The power output of solar PV can be expressed by the following expression in Eq. 2:

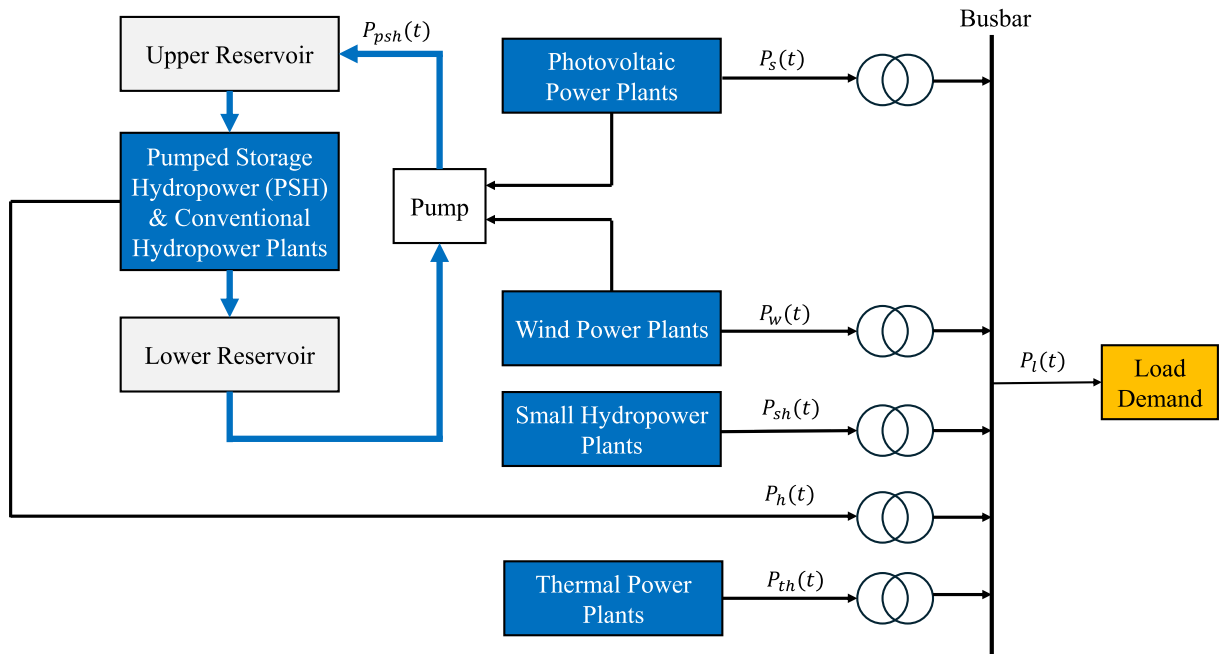


Fig. 1. Schematic Diagram of the Hybrid Solar-Wind-PSH system.

$$P_s(t) = IC_s \times PR \times \frac{POA(t)}{POA_{STC}} \quad (2)$$

where P_s is the total solar power output (W); IC_s denotes the solar installed capacity (MW). The solar power output from Eq. 2 is normalized by its maximum output to be obtained on a per-MW basis. The actual solar power output is then derived by the optimized solar installed capacity, IC_s , whose optimal value is determined through the capacity expansion optimization described in Section 3. POA is the plane of array irradiance incident on the solar PV at each time t (W/m^2). POA_{STC} represents the plane of array irradiance at Standard Test Conditions (STC) [43]. Typically, $1000 W/m^2$ which is used as the reference irradiance. PR denotes the Performance Ratio of solar PV array measuring the ratio of the actual power output to the theoretically maximum power output following IEC 61724-1 [44]. The commercial utility-scale PV can achieve PR of 0.9 under real operating conditions, as described by Vázquez et al. [45]. Therefore, this study assumes to be $PR=0.9$ to represent the high-efficiency of solar PV.

These three main components for POA are described as follows:

$$POA(t) = I_{dir}(t) + I_{diff}(t) + I_{gr}(t) \quad (3)$$

where:

1) The beam irradiance (I_{dir})

The beam irradiance is direct solar radiation without attenuation of diffusion and measured perpendicular to solar array. The beam irradiance can be expressed as:

$$I_{dir}(t) = DNI(t) \times \cos(AOI(t)) \quad (4)$$

where DNI is direct normal irradiance (W/m^2); AOI is the angle of incidence of beam radiation on a surface ($^\circ$), which is the angle between the beam radiation on the surface and the normal line to the surface. The AOI is changing with time and depending on the position of the sun changing during the day. The AOI can be expressed as follows:

$$\cos(AOI(t)) = \cos\theta_z(t)\cos\beta + \sin\theta_z(t)\sin\beta\cos(\gamma - \gamma_s(t)) \quad (5)$$

where θ_z is the zenith angle, which is the angle between the vertical to the normal line and sun; β is the slope of tilted angle ($^\circ$), which is the angle between the plane of the surface and the horizontal plane; γ_s is the solar azimuth angle ($^\circ$), which is the angular displacement from south of the projection of beam radiation on the horizontal plane; γ is the surface azimuth angle ($^\circ$). In Thailand, which is located in the northern hemisphere, PV panels are typically mounted facing south to maximize solar energy capture. Therefore, in this study, the tilted slope angle β is assigned to a constant value of 15° , while the surface azimuth angle γ is set to 0° .

2) The sky diffuse irradiance (I_{diff})

The sky diffuse irradiance is defined as the solar radiation reaching the earth's surface after having been scattered from the direct solar beam. The simplified Perez diffuse irradiance model for $\theta_z \leq 87.5^\circ$, proposed by Perez et al. [42] is given by

$$I_{diff}(t) = DHI(t) \left[(1 - F_1) \left(\frac{1 + \cos\theta_z(t)}{2} \right) + F_1 \left(\frac{a(t)}{b(t)} \right) + F_2 \sin\theta_z(t) \right] \quad (6)$$

$a(t)$ and $b(t)$ are two terms describing the incidence-weighted solid angle sustained by the circumsolar region as seen respectively by the tilted surface and the horizontal. In this form, the terms $a(t)$ and $b(t)$ of Eq.6 can be simply expressed as following equations

$$a(t) = \max(0, \cos(AOI(t))) \quad (7)$$

and

$$b(t) = \max(\cos 85^\circ, \cos\theta_z(t)) \quad (8)$$

The lower limit in the formulation of $a(t)$ and $b(t)$ are introduced to avoid possible distortions and to allow adequate model utilization for zenith angles higher than the validation domain of Perez model's study [46].

Given by Perez et al. [42,46] are a result of the approximation by assuming that all circumsolar energy originates from a point source. DHI represents diffuse horizontal irradiance (W/m^2); F_1 and F_2 are the Perez model circumsolar anisotropy coefficients and the Perez model horizontal/zenith anisotropy coefficients, respectively. The approximation of F_1 and F_2 is followed by Perez et al. [42,46].

The simplified Perez diffuse irradiance model for $87.5^\circ \leq \theta_z \leq 90^\circ$ [42] is given by

$$I_{diff}(t) = DHI(t) \left(\frac{1 + \cos\theta_z(t)}{2} \right) \quad (9)$$

3) The ground reflect irradiance (I_{gr})

The ground reflect irradiance is depending upon the irradiance on the ground according to the assumptions [42,47]. The array is infinitely long, Irradiance on the ground is uniform and equal to, i.e., horizon blocking and near-field shading by the array are ignored. Irradiance reflects from the ground equally in all directions, i.e., the ground is a diffuse or Lambertian reflector. The ground is visible to the array from the point of intersection of the array's slope projected to the ground, to the infinite horizon.

The ground reflect irradiance can be expressed as follows:

$$I_{gr}(t) = GHI(t) \times albedo \times \left(\frac{1 + \cos\beta}{2} \right) \quad (10)$$

where GHI is a global horizontal irradiance (W/m^2); $albedo$ is the diffuse reflection of solar radiation from the ground, and can be expressed as follows:

$$GHI(t) = DHI(t) \times DNI(t) \bullet (\cos\theta_z(t)) \quad (11)$$

Meteorological data, obtained from the Thai Meteorological Department [48], include daily solar radiation in 2023 records from monitoring stations across all provinces. These data are used to evaluate the spatiotemporal availability of solar resources.

2.2.2. Mathematical model for evaluating the wind power

The hourly wind power output is calculated using the average wind speed of each hour. Wind speed and wind turbine specifications are essential parameters for estimating power output. In this study, the wind turbine power output is modeled as a function of wind speed by applying the power curve characteristics as described by Gan et al. [37]. The power output of a wind turbine for different wind speed regions is expressed as:

$$P_w(t) = \begin{cases} 0 & \text{if } v \leq v_{cut-in} \text{ or } v > v_{cut-out} \\ IC_w \times \left(\frac{v(t)^3 - v_{cut-in}^3}{v_{rated}^3 - v_{cut-in}^3} \right) & \text{if } v_{cut-in} \leq v \leq v_{rated} \\ IC_w & \text{if } v_{rated} < v \leq v_{cut-out} \end{cases} \quad (12)$$

where P_w and IC_w denote the total wind power output and wind installed capacity specified by turbine's rated power output, respectively. The parameters v_{cut-in} , v_{rated} and $v_{cut-out}$, are the cut-in, rated and cut-out wind speeds. In this study, a representative wind turbine is selected with a rated capacity of 3 MW, and cut-in, rated, and cut-out wind speeds of 4 m/s, 15 m/s, and 25 m/s, respectively [49]. The actual wind power output is then derived by the optimized wind installed capacity,

IC_w , whose optimal value is determined through the capacity expansion optimization described in **Section 3**.

Meteorological data, obtained from the Thai Meteorological Department [48], include tri-hourly wind speed records in 2023 from monitoring stations across all provinces. These data were interpolated using linear interpolation to generate hourly wind speed values. Subsequently, the hourly wind speed records from all provincial stations were averaged to derive mean wind speed profile in national country.

2.2.3. Mathematical model for evaluating the hydro and PSH power

The power output of PSH for generating mode and pumping mode at time t can be expressed as **Eq. 13** and **Eq. 14**, respectively. Both formulations are derived from the hydropower and PSH operation model proposed by **Liu et al. [34]**

$$P_h(t) = \eta_{gen} g \rho Q_{gen}(t) \Delta H \quad (13)$$

$$P_{psh}(t) = \frac{g \rho Q_{pump}(t) \Delta H}{\eta_{pump}} \quad (14)$$

where P_h is the power generation of Hydropower plants (W) while P_{psh} is the pumping power of the PSH in Watt (W); g is gravitational acceleration, set to 9.81 m/s^2 , η_{gen} and η_{pump} is the generating efficiency and pumping efficiency; Q_{gen} and Q_{pump} are the volumetric flow rate for discharge generating water and for pumping water (m^3/h), respectively; ρ is the density of water, set to 1000 kg/m^3 , and ΔH is net water head height (m.).

In this study, the net head height ΔH and efficiency η were not employed directly as in **Eq. 13** and **Eq. 14**. Instead, the calculation utilized the hourly time series of generation output and water release for both generation and pumping operations to derive the average water rate. The average water rate for generation and pumping were calculated as follows:

$$\overline{WR}_{gen,i} = \frac{\sum Q_{gen,i}(t)}{\sum P_{h,i}(t)} \quad (15)$$

$$\overline{WR}_{pump,i} = \frac{\sum Q_{pump,i}(t)}{\sum P_{psh,i}(t)} \quad (16)$$

where $\overline{WR}_{gen,i}$ is defined as the volumetric water discharge released through the turbines per unit of electrical energy generated (m^3/MWh), representing the hydraulic requirement for electricity production of the hydropower plant i^{th} . Conversely, $\overline{WR}_{pump,i}$ is defined as the volumetric water flow rate per unit of electrical energy consumed by the pumping units (m^3/MWh), representing the water–energy conversion efficiency

of the pumping process of the PSH i^{th} . In this research, the average water rates for both power generation and pumping were calculated using actual operational data from 2012 to 2024 [39]. Storage capacity and average water rate of the twelve major EGAT hydropower and PSH are presented in **Table 2**. Therefore, **Eq. 13** and **Eq. 14** become:

$$P_{h,i}(t) = \frac{Q_{gen,i}(t)}{\overline{WR}_{gen,i}} \quad (17)$$

$$P_{psh,i}(t) = \frac{Q_{pump,i}(t)}{\overline{WR}_{pump,i}} \quad (18)$$

In addition to power evaluation, it is essential to account for the storage volume at each time step to ensure that it remains within the specified lower and upper limits of reservoir storage volume throughout the simulation period. The storage volume of the i^{th} reservoir is determined using the water balance equation followed by **Liu et al. [34]**, expressed as follows:

$$V_i(t+1) = V_i(t) + [(I_i(t) - D_i(t) + Q_{pump,i}(t) - Q_{gen,i}(t)) \times \Delta t] \quad (19)$$

where V_i is the storage volume in the reservoir (m^3); I_i is the natural flow from river which is the reservoir inflow (m^3/h); D_i is the spillway discharge (m^3/h); Δt is the time duration of time step (h). For PSH, a small amount of natural flow from rivers is required for supplementing evaporation and leakage losses, which is negligible in the operation. The reservoir inflows of 12 major EGAT hydropower plants were obtained from a long-term hydrological data from 2012 to 2024, collected by EGAT [50].

2.2.4. Levelized cost of electricity (LCOE) evaluation

LCOE is an index which measures the net present value of all costs over the lifetime of the asset divided by the discounted stream of the electricity output of the asset. This method relates the total energy generation cost with the total energy produced during the powerplant lifespans [51]. LCOE indicates how much electricity benefit must be generated compared to total investment cost. It has been widely used by researchers to evaluate Solar and/or Wind System configurations. However, LCOE formulation used in this study is based on a full Discounted Cash Flow (DCF) method rather than an annualized Capacity Recovery Factor (CRF) method. All yearly costs are discounted individually, allowing the model to incorporate the total annualized initial capital investment and the total annualized O&M costs. Conversely, the CRF method assumes a constant annualized cost, which is unsuitable for VRE technologies with multiple variable output. LCOE formulation developed from **Ganguly et al. [52]** is presented in **Eq. 20–Eq. 25**, and the economic parameters showed in **Table 3**.

Table 2

Storage capacity and average water rate for generation and pumping of twelve major EGAT hydropower plants and PSH.

Project Name	Storage Capacity (MCM)		Operating Mode	Total Operating Hours (h)	Average Water Rate $\overline{WR}$ (m^3/kWh)	Data Collection Year
	Lower Limit	Upper Limit				
Lamtakong Jolabha Vadhana	2.70	10.33	Generating	2180	1.55	2022–2024
			Pumping	3842	0.408	
Srinagarind Dam	10265.00	17745.00	Generating	67355	3.79	2012–2024
			Pumping	5036	7.154	
Bhumibol Dam	3800.00	13462.00	Generating	55840	3.92	2012–2024
			Pumping	277	18.449	
Rajjaprabha Dam	962.00	5638.84	Generating	54073	6.06	2012–2024
Vajiralongkorn Dam	2259.00	8860.00	Generating	66602	6.54	2012–2024
Bang Lang Dam	276.28	1454.36	Generating	54073	6.73	2012–2024
Sirikit Dam	2850.00	9510.00	Generating	60600	6.14	2012–2024
Ubol Ratana Dam	381.57	2931.30	Generating	58667	26.02	2016–2024
Sirindhorn Dam	831.38	1966.47	Generating	25205	11.85	2017–2024
Chulabhorn Dam	37.22	163.75	Generating	20443	1.54	2017–2024
Nam Pung Dam	1.51	23.23	Generating	24386	4.47	2017–2024
Huai Kum Dam	8.68	165.48	Generating	29702	15.3	2017–2024

Table 3

Main Economic Parameters of each power plant type.

Power Plant Types	Specific Investment Cost, α (THB/kW)	Lifetime, L (years)	Net Plant Heat Rate, HR (BTU/kWh)	Fuel Cost, FC (THB/M.BTU)
Solar PV [53,54]	34098.50	25	–	–
Wind [53]	47400	25	–	–
Small-Hydropower [55]	58846	25	–	–
Hydropower [53]	73865	60	–	–
PSH [53,56]	37525	60	–	–
Thermal [57]	41116	25	6800	10483

$$LCOE = \frac{TAC}{E_{served}} \quad (20)$$

where TAC is the total annualized cost, and E_{served} is the total annualized energy generation for all types of power plants supply to load demand. The total annualized cost (TAC) is shown in Eq. 20

$$TAC = C_{capital} + C_{O\&M} + C_{Fuel} \quad (21)$$

where TAC denotes the total annualized capital cost, includes annualized initial capital costs $C_{capital}$, annualized total Operation and Maintenance (O&M) costs $C_{O\&M}$ for Solar PV, Wind, Hydropower, PSH, and Thermal power plants. The total annualized fuel costs C_{Fuel} considered only for the thermal power plants. The annualized initial capital cost ($C_{capital}$) is shown in Eq. 21

$$C_{capital} = \alpha_s IC_s + \alpha_w IC_w + \alpha_{sh} IC_{sh} + \alpha_h IC_h + \alpha_{psh} IC_{psh} + \alpha_{th} IC_{th} \quad (22)$$

where $C_{capital}$ represent the annualized initial capital cost of power plant type. In this formulation, $\alpha_s, \alpha_w, \alpha_{sh}, \alpha_h, \alpha_{psh}, \alpha_{th}$ denotes the specific investment cost (THB/kW) of the solar PV, wind, small hydropower, hydropower, PSH, and thermal power plant respectively, while $IC_s, IC_w, IC_{sh}, IC_h, IC_{psh}, IC_{th}$ are the installed capacity (MW) of the respective power plant type. The total annualized O&M costs ($C_{O\&M}$) is shown as follows:

$$C_{O\&M} = \frac{OM_s}{(1+i)^{L_s}} + \frac{OM_w}{(1+i)^{L_w}} + \frac{OM_{sh}}{(1+i)^{L_{sh}}} + \frac{OM_h}{(1+i)^{L_h}} + \frac{OM_{psh}}{(1+i)^{L_{psh}}} + \frac{OM_{th}}{(1+i)^{L_{th}}} \quad (23)$$

where $C_{O\&M}$ represents the total annualized O&M costs, including Fixed O&M Costs and Variable O&M Costs; $OM_s, OM_w, OM_{sh}, OM_h, OM_{psh}, OM_{th}$ are the total O&M cost of each power plant types, which are approximated to be 2% of the initial investment $C_{capital,j}$ for each power plant type as given by Canales et al. [10]. i is the discount rate, which is specified as 5.26%; $L_s, L_w, L_{sh}, L_h, L_{psh}, L_{th}$ are the power plant lifetime for each power plant type in years. The total annualized fuel costs (C_{Fuel}) is shown as follows:

$$C_{Fuel} = \frac{1}{(1+i)^{L_{th}}} \sum_{t=1}^{8760} P_{th}(t) \cdot \Delta t \cdot HR \cdot FC \quad (24)$$

where C_{Fuel} represents the total annualized fuel costs for thermal power plants over one year; FC is the fuel cost of thermal power plants. The total annualized energy generation E_{Served} is shown as follows:

$$E_{Served} = \sum_{t=1}^{8760} P_i(t) \cdot \Delta t \quad (25)$$

where E_{Served} represents the total annualized energy generation for all types of power plants supply to load demand. The economic parameters used in Eq. 20 – Eq. 25 are summarized in Table 3.

2.3. Model validation

To verify the reliability of the proposed coordinated dispatch model developed in this study, three validations were conducted: (1) storage volume of hydropower plant water balance model, (2) hydropower plants power dispatch model, and (3) solar power generation model.

2.3.1. Validation of storage volume of hydropower plant water balance model

The validation of reservoir storage volume was performed by comparing the simulated hourly storage profiles with the actual hourly storage data in 2023 provided by EGAT [50]. The simulated storage volume was computed using Eq. 18, which incorporates inflow, spillway discharge, release for power generation, and release for pumping from the actual hydrological data [50]. This calculation yielded an hourly simulated storage volume that was compared against historical data from 12 EGAT hydropower plants in 2023. The monthly storage volume profiles showed good agreement with actual records, as illustrated in Fig. 2, confirming the model's ability to accordance with practical operational characteristics. Most plants exhibit the same trend, which was successfully captured by the model.

In addition, the water balance model accuracy was also quantified using the Mean Relative Absolute Error (MRAE). MRAE is employed to account for both errors and absolute relative magnitude errors of hourly storage volume between the simulated and the actual values. The MRAE of the storage volume of hydropower plant i^{th} is expressed as:

$$MRAE_{V,i} = \frac{1}{8760} \sum_{t=1}^{8760} \left(\frac{|V_{i,model}(t) - V_{i,actual}(t)|}{V_{i,actual}(t)} \right) \quad (26)$$

where $V_{i,model}(t)$ and $V_{i,actual}(t)$ denote the simulated and actual hourly reservoir storage volumes of hydropower plant i^{th} , respectively.

The validation results in Table 4 indicate that the simulated reservoir storage volumes show strong agreement with the actual data, with MRAE values generally below 1% for most hydropower plants. This confirms that the water-balance formulation used to compute water release for generation, and water release for pumping is highly consistent with actual operations.

2.3.2. Validation of hydropower plants power dispatch model

The validation of hydropower plants power dispatch model was performed by comparing the simulated hourly hydropower dispatch profiles with the actual data from 2012 to 2024 provided by EGAT [39]. The model accuracy of hydropower plants power dispatch model for both generating and pumping modes of hydropower plant i^{th} were quantified using the MRAE as:

$$MRAE_{gen,i} = \frac{1}{n_{gen,i}} \sum_{t=1}^{n_{gen,i}} \left(\frac{|P_{gen,model,i}(t) - P_{gen,actual,i}(t)|}{P_{gen,actual,i}(t)} \right) \quad (27)$$

and

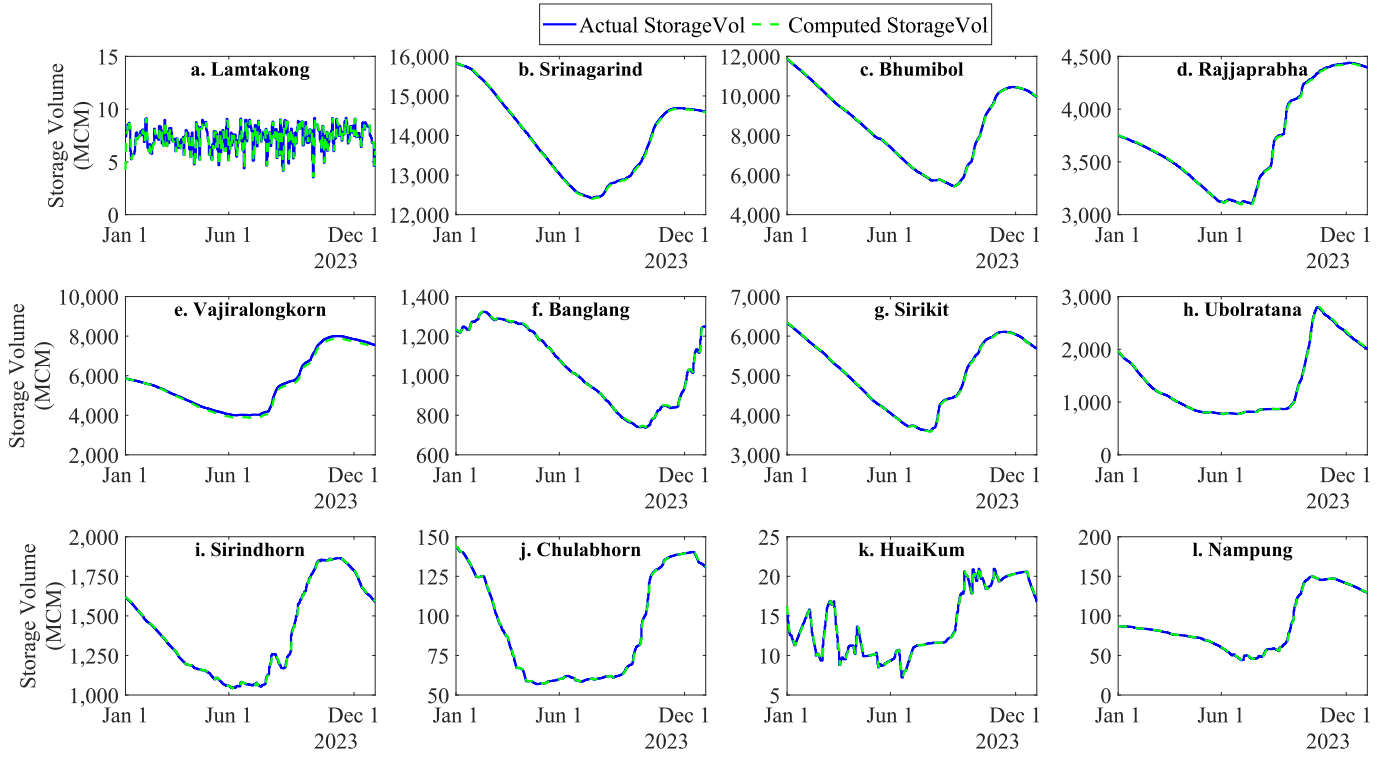


Fig. 2. Comparison of Actual and Computed Storage Volume in 2023.

(a) Lamtakong, (b) Srinagarind, (c) Bhumibol, (d) Rajjaprabha, (e) Vajiralongkorn, (f) Banglang, (g) Sirikit, (h) Ubolratana, (i) Sirindhorn, (j) Chulabhorn, (k) Huai Kum, (l) Nampung.

Table 4

MRAE of Storage Volume, Hydropower Plants Dispatch, and Solar Energy.

Project Name	MRAE (%)			
	Storage Volume	Operating Mode	Hourly Energy	Solar Energy
Lamtakong	2.373	Generating Mode	3.249	–
Srinagarind	0.122	Pumping Mode	2.151	9.812
		Generating Mode	2.623	
Bhumibol	0.198	Pumping Mode	3.026	9.812
		Generating Mode	3.714	
Rajjaprabha	0.166	Pumping Mode	3.069	9.683
		Generating Mode	1.757	
Vajiralongkorn	1.842	Generating Mode	2.359	9.819
Banglang	0.258	Generating Mode	1.757	9.665
Sirikit	0.216	Generating Mode	1.816	9.829
Ubolratana	0.444	Generating Mode	3.235	10.451
Sirindhorn	0.198	Generating Mode	3.385	9.844
Chulabhorn	0.277	Generating Mode	3.336	9.791
Huai Kum	0.897	Generating Mode	4.957	9.794
Nampung	0.330	Generating Mode	3.045	9.842

$$MRAE_{pump,i} = \frac{1}{n_{pump,i}} \sum_{t=1}^{n_{pump,i}} \left(\frac{|P_{pump,model,i}(t) - P_{pump,actual,i}(t)|}{P_{pump,actual,i}(t)} \right), \quad (28)$$

respectively. $P_{i,model}(t)$ and $P_{i,actual}(t)$ represent simulated and actual power outputs of power plant i^{th} , respectively, and n_i denotes the

number of operating hours of each hydropower plant i^{th} as shown in Table 2. The subscript gen and $pump$ denote to the generating mode and pumping mode of operations, respectively.

From Table 4, the generating-mode MRAE values range from 1.7% to 4.9%, while the pumping-mode errors remain within 2–3% for most PSH plants. These low error levels demonstrate that the coordinated dispatch framework accurately captures both generation and pumping behaviors under real operating conditions.

2.3.3. Validation of solar power generation model

The validation of solar power generation was performed by comparing the simulated solar outputs calculated in Section 2.2.1 with the Global Solar Atlas [58]. The simulated model operates using a full-year dataset comprising of 8760 hourly time steps, while the Global Solar Atlas provides only monthly average hourly PV profiles. Therefore, the actual total monthly solar PV power output in each month was allocated to each dam according to its geographical location, and the simulated hourly results were reformatted into the same monthly average–hourly structure. The actual dataset from the Global Solar Atlas represents the mean solar PV power output for each hour of the day across each month. The simulated model was rearranged into the similar format of the dataset from the Global Solar Atlas to validate the results using MRAE for solar PV as following equation:

$$MRAE_s = \frac{1}{n_s} \sum_{t=1}^{n_s} \left(\frac{|P_{s,model}(t) - P_{s,GSA}(t)|}{P_{s,GSA}(t)} \right) \quad (29)$$

where $P_{s,model}(t)$ and $P_{s,GSA}(t)$ represent the simulated and actual hourly solar PV power outputs, respectively. n_s denotes a number of average hourly solar irradiance availability.

From Table 4, The solar PV validation also shows satisfactory performance, with MRAE values below 1% across stations. The deviations are close to the monthly Global Solar Atlas dataset. This confirms that the simulated solar power output is highly consistent with the monthly

Global Solar Atlas dataset.

Overall, the validation confirms that the proposed model reliably represents the reservoir dynamics, hydropower dispatch in both operating modes, and solar PV power generation, thereby ensuring confidence in optimization framework.

2.4. Scheduling operation optimization methods

This study investigates the optimal increase in the installed capacities of solar and wind, along with the evaluation of optimal dispatch strategies under the existing EGAT's hydropower and PSH system, aiming to maximize the feasible RE share while minimizing the LCOE. To achieve this research objective, a hybrid scheduling strategy was developed based on the coordinated operation of 12 major EGAT-operated hydropower and PSH plants. The dispatch hierarchy prioritized solar and wind generation, followed by hydropower constrained by reservoir and water allocation limits. Thermal power is utilized only as a last resort to meet any residual unmet demand.

The proposed framework begins with assigning weights to each objective component, such as maximizing RE share, minimizing power shortfalls, reducing generation variability, and lowering the LCOE. These weight factors allow for dynamic prioritization of objectives based on planning scenarios. The objective function B_{Total} is defined mathematically as follows:

$$\max B_{Total} = w_{B_1}B_1 + w_{B_2}B_2 + w_{B_3}B_3 + w_{B_4}B_4 \quad (30)$$

where B_n is represented the benefit value of indicator n , and w_{B_n} is represented the weight assigned to benefit B_n . Each benefit function is evaluated independently, including: (1) maximizing RE share, (2) minimizing the LCOE, (3) maximizing hydropower output, and (4) mitigating thermal power output fluctuation. Lists of each benefit function shown in Table 5.

To maximize RE shares served as the primary objective, emphasizing the integration of variable renewables into the national grid. Complementary objectives included minimizing LCOE to ensure economic viability, Hydropower Potential Utilization to assess the full operational capacity of existing hydropower plants and Contracted Thermal Capacity Stabilization to maintain grid stability by reducing thermal power fluctuations.

In the multi-objective optimization framework, the primary objective was to maximize the RE share. However, multiple cases are considered simultaneously, each with distinct priorities. To address this, weight factors (w_{B_n}) were assigned to different objectives. A higher weight on RE share, for example, increased the likelihood of solutions favoring high RE share, resulting in outcomes that emphasize renewable integration. Conversely, assigning higher weights to other objectives would shift the optimization focus accordingly.

The constraints were implemented to formulate the operational

Table 5
Summary Technical and Operational Benefit Function.

The benefit indicators B_n	Formulation	w_{B_n}
B_1 : Renewable Energy Share (RE Share)	$B_1 = \frac{\sum_t [P_s(t) + P_w(t) + P_{sh}(t) + \sum_i (P_{hi}(t) - P_{psi}(t))] \cdot \Delta t}{\sum_t (P_i(t) \cdot \Delta t)}$	10
B_2 : Levelized Cost of Electricity (LCOE)	$B_2 = \frac{LCOE_{ref}}{LCOE}$ where $LCOE_{ref} = 1.37$ THB/kWh	1
B_3 : Hydropower Potential Utilization	$B_3 = \frac{\sum_t \sum_i P_{hi}(t)}{8760 \times \sum_i P_{hi}(t)}$	1
B_4 : Contracted Thermal Capacity Stabilization	$B_4 = 1 - \frac{std.P_{th}(t)}{P_{th}(t)}$	1

requirements and limitations such as the total generation units, associated with the solar, wind, thermal, and PSH power generation were identified, as shown in Table 6.

2.4.1. Compressed sensing (CS) for dimensionality reduction

The optimization process began with the construction of a high-dimensional decision vector comprising 105122 variables, which include hourly dispatch schedules for 12 major EGAT hydropower plants (each represented by an 8760-h profile), and scalar decision variables for the installed capacities of solar and wind power. Due to the extreme dimensionality, a CS technique was employed to project the original decision space into a compressed domain while preserving the essential characteristics of the system. This dimensionality reduction served a dual purpose: it significantly reduced the computational complexity of the optimization task and enhanced convergence efficiency by concentrating the search on the most informative features. In this study, the original 105122-dimensional vector was effectively compressed into approximately 1155 variables.

Li et al. [60] studied the economic dispatch with high RE share. They encountered severe challenges due to extremely high-dimensional decision spaces—often in the order of tens of thousands of variables (e.g., hourly dispatch across multiple assets over extended periods). The dimensional problem of scheduling operation caused by optimizing the hourly generation amount could be relaxed due to enhanced both computational tractability and convergence performance in the scheduling problem. Most previous studies [61–63] employed various strategies. However, while conventional approaches such as Principal Component Analysis (PCA) [61,62], or autoencoder-based reductions [63] showed promise, they often failed to fully capture the underlying sparsity of the decision space in large-scale RE systems. To overcome these limitations, this study explores CS [64–67] as a theoretically grounded dimensionality reduction framework.

In large-scale energy system optimization problems, particularly those involving hourly dispatch across multiple renewable and storage assets, the dimensionality of the decision space often becomes prohibitively high (e.g., >100000 variables). To address this challenge, we extend ideas from the field of CS, the communication of intelligent network, to employ as an effective dimensionality reduction technique that retains the essential structure of the solution while significantly reducing computational burden. The CS formulation adapted from Eldar and Kutyniok [64].

In the proposed framework, the full-dimensional hourly dispatch decision vector of 12 EGAT's hydropower plants over a one-year horizon with 105122 dimensions is first assumed to exhibit sparsity or approximate sparsity in a known transform domain defined by a basis matrix Ψ . This assumption allows the representation of x as follow:

$$x = \Psi\theta \quad (43)$$

Here, Ψ represents as the sparse basis, transforming the original hourly hydropower dispatch variables into a domain where most coefficients are zero or near zero; θ represents the sparse coefficient vector in the transform domain.

To reduce the dimensionality of the hydropower dispatch optimization problem, a sensing matrix $\Phi \in \mathbb{R}^{k_{CS} \times n_{ix}}$, where k_{CS} denotes the compressed dimension in the compressed domain after applying the dimensionality reduction while n_{ix} represents the total number of hourly hydropower dispatch variables across 12 EGAT hydropower plants over a one-year horizon. and $k_{CS} \ll n_{ix}$, is applied to the full-dimensional hourly dispatch decision vector x , yielding a compressed representation as follows:

$$y = \Phi x = \Phi \Psi \theta \quad (44)$$

where y denotes the measurement vector in the compressed domain. The sensing matrix is designed to satisfy the Restricted Isometry Property (RIP), ensuring incoherence between the measurement and sparsity

Table 6

Summary of Operational Constraints for Hybrid Renewable Energy Systems.

Constraint Category	Mathematical Expression	Equation Number	Description
Solar Power Generation	$P_s^{min}(t) \leq P_s(t) \leq P_s^{max}(t)$	(31)	Lower and upper limit of solar power, which is supplying to the electricity load demand at time period t
Wind Power Generation	$N_s^{min}(t) \leq N_s(t) \leq N_s^{max}(t)$	(32)	Limits on the number of solar PV modules at time period t
Hydro and PSH Power Generation	$P_w^{min}(t) \leq P_w(t) \leq P_w^{max}(t)$	(33)	Lower and upper limit of wind power at time period t
	$P_{hi}(t) \geq 0$	(34)	the hydro power supplying to load demand at time period t
	$P_{psh,i}^{min}(t) \leq P_{psh,i}(t) \leq P_{psh,i}^{max}(t)$	(35)	Lower and upper limit of power output for pumping power at time period t
	$P_{hi}^{min}(t) \leq P_{hi}(t) \leq P_{hi}^{max}(t)$	(36)	Lower and upper limit of power output for generating power at time period t
Storage Volume Constraints	$V_i^{min}(t) \leq V_i(t) \leq V_i^{max}(t)$	(37)	Lower and upper limits of reservoir storage volume at time period t
	$V_i^{max}(t) - V_i^{min}(t) \leq V_{cap,i}$	(38)	Difference between maximum and minimum storage volume constrained by capacity
Thermal Power Generation	$V_i^{min}(t) \leq V_{final,i}(t) - V_{initial,i}(t) \leq V_i^{max}(t)$	(39)	The difference between the initial and final storage volume over period time t
	$P_{th}(t) = P_l(t) - [P_s(t) + P_w(t) + P_h(t) - P_{psh}(t)]$	(40)	Thermal power fills residual load demand at time period t
Power Balance	$P_l(t) = P_s(t) + P_w(t) + P_{th}(t) + P_h(t) + P_{sh}(t) - P_{psh}(t)$	(41)	Power balance equation ensuring demand is satisfied at time period t
Reliability Constraints	$LOLE = \frac{1}{8760} \sum_{t=1}^{8760} \delta(t)$	(42)	System reliability limit
	$\delta(t) = \begin{cases} 1 & \text{if } P_l(t) > P_{Gen}(t) \\ 0 & \text{otherwise} \end{cases}$		where $P_{Gen}(t)$: Total generation from $P_s + P_w + P_{sh} + P_h - P_{psh} + P_{th}$ at time period t
	Constraint : $LOLE \leq LOLE_{max}$		$LOLE_{max}$: 16.8 h/year [59]

domains for accurate recovery.

The optimization process is conducted directly in the compressed domain y , where the Hybrid Genetic Algorithm–Particle Swarm Optimization searches for an optimal solution y^* . Once an optimal compressed solution is identified, reconstruction of the original high-dimensional hydropower dispatch x^* is achieved by solving via ℓ_1 -norm minimization as follow:

$$\min \|\theta\|_1 \text{ subject to } y^* = \Phi \Psi \theta \quad (45)$$

This approach promotes sparsity in θ while ensuring the reconstructed vector satisfies the measurement constraint. The recovered vector is then computed as follows:

$$x^* = \Psi z^* \quad (46)$$

After the optimization process is complete, the optimal hourly hydropower dispatch decision variables z^* obtained from the hybrid GA–PSO optimization in the compressed domain.

In the CS-based formulation of this study, the original dimension space x of size 105122 was reduced to a reduced-dimensional space z^* of size 1155. The original position vector x^* is reconstructed from z^* using the sensing matrix Φ . The objective functions, Eq. 29, based on x and x^* were calculated and compared. The comparison results indicate that the dimensionality reduction yields an approximation drift below 5%. Therefore, the proposed computational algorithm proved to be as efficient as it was robust, handling large datasets with ease while maintaining high accuracy in the results.

2.4.2. Hybrid genetic algorithm – Particle swarm optimization for scheduling operation

To further enhance global and local search capabilities, this study adopted a hybrid Genetic Algorithm – Particle Swarm Optimization (GA–PSO) approach. The GA procedure developed from the GA framework of Liu et al. [34], while PSO procedure is based on the theory presented in Particle Swarm Optimization's theory [68]. The rationale behind this integration lies in the complementary strengths of the two algorithms: GA provides robust global exploration through evolutionary operators such as selection, crossover, and mutation, while PSO offers rapid local convergence by simulating social behaviors among particles.

The hybrid structure illustrates in Fig. 3, which presents the iterative framework used in this study. Initially, a population is initialized using GA, where individuals encode decision variables for hydropower

dispatch and the installed capacities of solar and wind power. The proposed optimization framework employed a GA operating within a compressed domain, and evolutionary operations including crossover, and mutation were conducted in the compressed domain based on the objective function (i.e., the objective function described in Section 2.4).

The offspring generation, denotes the population of new solutions generated from parent individuals in GA, is conducted using uniform crossover, wherein each decision variable is independently inherited from either of the two selected parents with equal probability shown as follows:

$$z_{offspring}^*(k) = \begin{cases} z_A^*(k), & \text{if } r_k < 0.5 \\ z_B^*(k), & \text{otherwise} \end{cases} \text{ for } k = 1, 2, \dots, k_{CS} \quad (47)$$

where $z_{offspring}^*(k)$ denotes the k^{th} decision variables of offspring in compressed domain; $Z_A(k)$ and $Z_B(k)$ are the k^{th} decision variables of the two parents; r_k is the random variable from ranging between [0,1].

Gaussian mutation operator is applied to offspring. Each decision variable undergoes random perturbation with probability, defined as follows:

$$z^*(k) = \begin{cases} z^*(k) + N(0, \sigma^2), & \text{if } q_k < \mu \\ z^*(k), & \text{otherwise} \end{cases} \text{ for } k = 1, 2, \dots, k_{CS} \quad (48)$$

where $N(0, \sigma^2)$ is a zero-mean Gaussian distribution, and μ is the mutation rate.

Elite selection was performed to ensure that the best individual, defined as the solution with the lowest LCOE and highest RE share, is preserved across generations. Subsequently, PSO was applied to refine the solutions in the compressed domain derived through CS. Each particle was evaluated for multi-objective search according to the objective function. Subsequently, the best position of each particle and the best position among all particles were calculated. In the CS formulation, the PSO algorithm operated in the compressed domain. The velocity and position updates in the compressed domain are formulated as follows:

$$v_{z,i}^{n_{iter}+1} = w_{PSO} \bullet v_{z,i}^{(n_{iter})} + c_1 r_1 (p_{z,i}^{best} - z_i^{(n_{iter})}) + c_2 r_2 (g_z^{best} - z_i^{(n_{iter})}) \quad (49)$$

where $z_i^{(n_{iter})}$ is the position of each particle in the current iteration i , $p_{z,i}^{best}$ is the best position of each particle, g_z^{best} is the position of the best particle, $v_{z,i}^{(n_{iter})}$ is the velocity in compressed domain of each particle at iteration n_{iter} , w_{PSO} is the learning factor according to the velocity of the

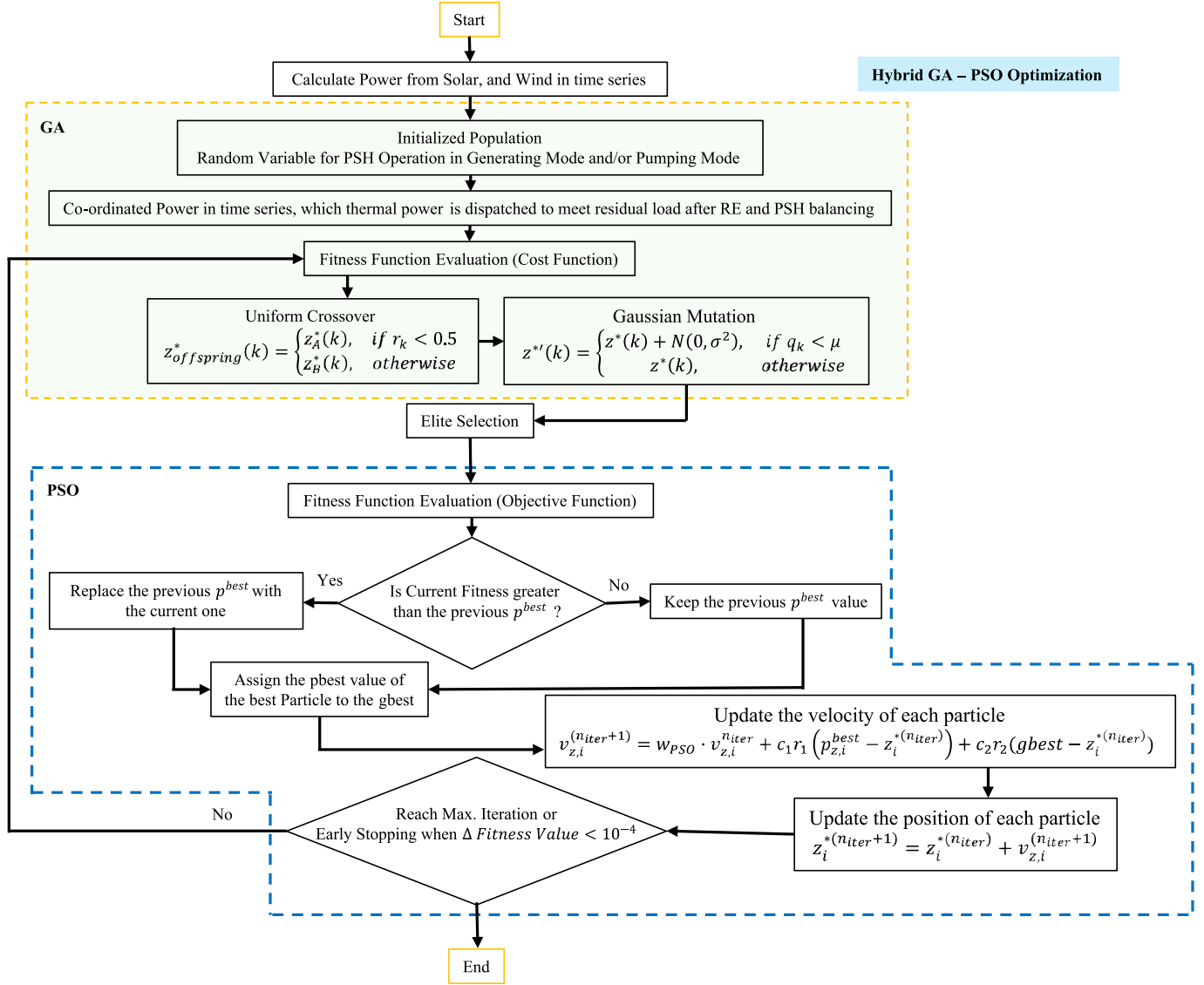


Fig. 3. Flowchart of Hybrid Genetic Algorithm – Particle Swarm Optimization Scheme.

current state, and c_1 and c_2 are the learning factors according to the best position of each particle and the position of the best particle, respectively. r_1 and r_2 are random variables uniformly distributed in $[0,1]$ between the particles, and $v_{z,i}^{n_{iter}+1}$, is the new velocity which determines the new position of each particle. The new position update in compressed domain is formulated as follows:

$$z_i^{*(n_{iter}+1)} = z_i^{*(n_{iter})} + v_{z,i}^{(n_{iter}+1)} \quad (50)$$

The formulation of Eq. 49 and Eq. 50 must take into account that $z_i^{*(n_{iter})}$, $p_{z,i}^{best}$, and $g_{z,i}^{best}$ are conformed with n_{ch} characteristics that belong to integer values ($n_{ch} \in \mathbb{Z}$). For this reason, the new velocity $v_{z,i}^{(n_{iter}+1)}$ as well as a new position $z_i^{*(n_{iter}+1)}$ in the compressed domain at iteration $n_{iter} + 1$ contain only integer values in their calculation.

Applying Eq. 49 and Eq. 50 for each particle, the new state of particles was obtained; this continued to be evaluated until the decision criteria, or the number of iterations were met. For this study, the number of particles p_{PSO} was 100, w_{PSO} was set to 1.5, c_1 , and c_2 were 2.5 and 3.5, respectively, and the number of iterations was 500.

After each update, the latent position $z_i^{*(n_{iter}+1)}$ is reconstructed to the original high-dimensional space using the sensing matrix Φ to

reconstruct the full-dimensional vector as follows:

$$x_i^{*(n_{iter}+1)} = \Phi z_i^{*(n_{iter}+1)} \quad (51)$$

where $x_i^{*(n_{iter}+1)}$ is represented the position vector of particle i at iteration $n_{iter} + 1$; Φ is represented the sensing matrix used for dimensionality reduction, and $z_i^{*(n_{iter}+1)}$ is represented the latent position vector of particle i at iteration $n_{iter} + 1$.

The PSO sub-process iterated over a fixed number of generations or until a convergence criterion is met, effectively fine-tuning the elite solutions from the GA stage. Once refined, these solutions were re-integrated into the GA population for the next generational loop.

To evaluate optimization performance, a compressed decision vector with 1155 variables was randomly initialized. These compressed parameters were then used to reconstruct full-dimensional decision variables, which were subsequently evaluated through the objective function and subject to the constraints, as described in Section 2.4.

The GA was configured with a population size was set at 100, the number of parents was set at 32, the generational crossing was made at uniform crossover, the mutation took a value of 2% of the total genes of the new individuals generated by the generational crossing, and the total number of iterations was 500.

After evaluating each individual in the multi-objective optimization, the best individuals were selected based on their minimized LCOE. The algorithm monitors the improvement of the best cost value across generations. If the relative change in total benefit falls below 10^{-4} for 10 consecutive generations, the GA process terminates, and the best individuals are passed to the PSO stage for further refinement. Subsequently, the PSO algorithm utilized the GA-derived best solutions as initial particles and performed iterative velocity and position updated to explore the solution space and improve convergence. This hybrid approach combined the global search capability of GA with the local exploitation ability of PSO, and the process continued until convergence criteria, or the maximum iteration limit was met.

Despite the approximation drift introduced by dimensionality reduction, the optimization performance remains unaffected. As discussed in the preceding section, the dimensionality reduction process resulted in an approximation drift of approximately 5%. This was attributed to the reconstruction of solutions into the full-dimensional space prior to benefit function evaluation, ensuring accurate the objective function assessments and reliable optimization outcomes. This process of reconstructing solutions into the original high-dimensional space allowed the optimization algorithm to evaluate the technical and economic benefits accurately.

3. Scenario designs for case study

This study investigates the maximum achievable RE share in Thailand's power system through the optimal operation of existing EGAT's hydropower and PSH. In this analysis, the PSH are primarily operated as ESS, while conventional hydropower plants are operated to maximize electricity generation without considering the water demands from the Royal Irrigation Department (RID). The key objective is to determine the maximum RE share that can be supported by fully leveraging the operational capabilities of existing hydropower plants and PSH to accommodate the highest possible installed capacities of solar and wind power.

Three scenarios were developed to evaluate power system dispatch strategies under varying installed capacity plans and optimization approaches. Scenario 1, **Actual EGAT's Installed Capacity and Operation** defined as the baseline scenario, which the installed capacities of EGAT's conventional hydropower plants and PSH at 3731.6 MW. The installed capacities of solar and wind were set accordance with the Power Development Plan (PDP) 2024 [59]. Noted that the average hourly load demand was derived from datasets during 2012–2024, as presented in Section 2.1. The wind speed and solar irradiance data for the year 2023 were obtained as described in Sections 2.2.1 and 2.2.2, respectively. Hydrological data—including reservoir inflow, turbine discharge, pumped storage volume, water losses, evaporation, irrigation

outlet, and spillway discharge—were also obtained from the average hourly historical records for 2012–2024 to ensure consistency across all datasets, as detailed in Section 2.2.3.

Scenarios 2 and 3 were designed to evaluate the improvement in RE integration to coordinated operation by optimizing the installed capacities of solar PV and wind. In both scenarios, the existing installed capacity of hydropower was used, while the hourly load demand and reservoir inflow were kept identical to those in Scenario 1. Nonetheless, the operational strategies for hydropower and PSH were optimized to maximize the RE share.

In Scenario 2, **Optimized Installed Capacity and Operation without utilizing PSH**, Optimization approach was used to evaluate the maximum optimal installed capacities of solar PV and wind power without RE curtailment, and dispatch hourly hydropower generation. The optimization aims to ensure that the integrated hourly solar PV and wind generation closely equivalent the load demand without PSH for system balance, exceeding the load demand would result in RE curtailment as shown the conceptual representation in Fig. 4(a).

In Scenario 3, **Optimized Installed Capacity and Operation with utilizing PSH**, employed the same optimization approach as Scenario 2 but further increases the installed capacities of solar PV and wind beyond the load demand. In this scenario, PSH was utilized to manage the excess generation from solar and wind by storing surplus energy through pumping operations during periods of low demand and releasing it for electricity generation during peak demand periods as illustrated in the conceptual representation in Fig. 4(b). This scenario highlights the critical role of PSH in effectively managing surplus RE and ensuring system balance under high renewable share conditions.

4. Optimal results and discussion

This scenario-based design establishes a foundation for analyzing the optimized dispatch of Thailand's hybrid RE systems, particularly focusing on the coordinated operation of 12 EGAT hydropower plants and the role of PSH in enabling higher RE shares. This section presents the power dispatch planning for hybrid energy systems under three distinct scenarios, as described in Section 3. Each scenario explores different levels of renewable energy integration and operational strategies, with a particular focus on hydropower and PSH systems operated by EGAT. The increase of optimal solar PV and wind installed capacities for each scenario induced distinct operational strategies of each power plant, indicated that differentiated dispatch patterns, generation and pumping time periods, and seasonal storage volume throughout the entire year. To examine the influence of thermal power availability on economic dispatch, the analysis compared various periods which affect the increasing of RE share. These comparisons underscored the critical role of PSH in mitigating renewable curtailment and maintaining system

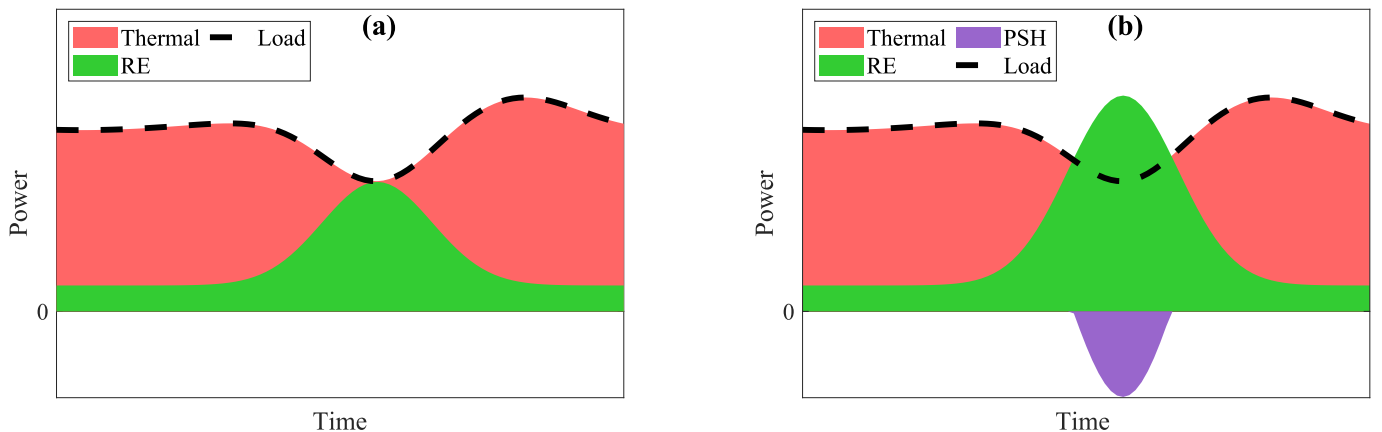


Fig. 4. Conceptual Representation of Optimized Scheduling Operation scenarios: (a) without utilizing PSH, (b) with utilizing PSH.

adequacy, particularly in the absence of dispatchable thermal capacity.

The optimized results determined optimized installed capacities of solar PV and wind for Scenario 2 and 3, alongside the corresponding optimal operating schedules. The discussion is structured in two parts: **Section 4.1** discusses the optimized dispatch operations under different load demand conditions and hydrological constraints, highlighting the temporal interplay between hydropower, PSH, and variable RE resources across three operational scenarios. **Section 4.2** discusses the optimum installed capacities of solar and wind power associated with each scenario, analyzing their impact on both the RE share and the overall system of LCOE.

4.1. Optimized dispatch under scenario-based analysis

The simulation was conducted for the entire year; however, two representative periods were selected to illustrate the scheduling dispatch operation: January 1 to January 7, corresponding to the lowest load demand period, and May 23 to May 29, representing the highest load demand period. In addition, the storage volume and the operation of hydropower and PSH throughout the year were also presented under three scenarios in this section.

Fig. 5 presents Scenario 1 load profiles over two representative periods: January 1 to January 7 for low load demand period and May 23 to May 29 for high load demand period. The results illustrated that thermal power plants remain the primary source of electricity generation with their power output being particularly higher during periods of high load demand than low load demand periods, while the current utilization of solar and wind energy remained limited, highlighting substantial potential for further expansion. Moreover, hydropower and PSH operations contributed relatively little to total generation because their scheduling operation is strongly constrained by downstream irrigation requirements governed by the Royal Irrigation Department (RID), particularly for agricultural use.

Fig. 6 illustrates the monthly variation in total storage volume of 12 major EGAT's reservoirs under coordinated hydropower and PSH operation from Scenario 1. The total storage volume exhibited a pronounced seasonal pattern, starting at a relatively high storage volume in January and progressively declined to its minimum in July–August due to continuous water releases for power generation. From August onwards, the total accumulated inflow increased, leading to a rapid recovery of storage volume and peaking in November–December. Hydropower

generation is continuously observed during January–July when storage volume remain high, drops substantially between August and October, and subsequently increases toward the end of the year following total accumulated inflow recovery. In contrast, pumping operations exhibit limited and temporally irregular patterns throughout the year. The highest pumping operations occur in May and October highlighting the underutilization of PSH operation.

In the actual EGAT operation, thermal power generation remains dominant. In contrast, the **Optimized Operation without PSH scenario** substantially reduced thermal reliance during high-RE periods. As illustrated in Fig. 7, during daytime hours when solar and wind contributions effectively reduced thermal reliance, hydropower compensated for the deficit in solar and wind generation by supplying electricity during nighttime periods when RE output was inadequate. During periods of low solar and wind power output during day, thermal power generation increased despite available stored water, while in peak demand periods, hydropower mitigated thermal generation when solar and wind output were insufficient. Nevertheless, the dispatch profiles during both low-demand and high-demand periods exhibited highly similar patterns, indicating an operational regime that does not materially differentiate by load demand.

Fig. 8 indicates that Scenario 2 exhibits a seasonal total storage volume of Scenario 2 follows a similar pattern to Scenario 1 as shown in Fig. 6. Storage volumes were high in the early months, underwent a gradual decrease toward mid-year, and subsequently recovered during the later months, driven by increased inflows. Nevertheless, the storage volume consistently remained within the lower and upper limits. In terms of energy production, hydropower generation with temporal variability was observed during January–July and November–December, decreased markedly from July to August due to declining storage volume, and slightly increased again toward the end of the year. By contrast, the unavailable utilization of PSH operation under the current hydrological and operational constraints.

High solar and wind penetration resulted in **Optimized Operation with PSH scenario from Scenario 3**, particularly during midday hours—leading to significant RE curtailment. Fig. 9 shows that daytime solar generation frequently exceeds load, especially in low-load periods. During daylight hours, the total output of solar, wind, and small hydropower was sufficient to meet load demand, eliminating the need for hydropower and thermal dispatch. By contrast, in the morning and nighttime periods—when solar output was unavailable and wind output

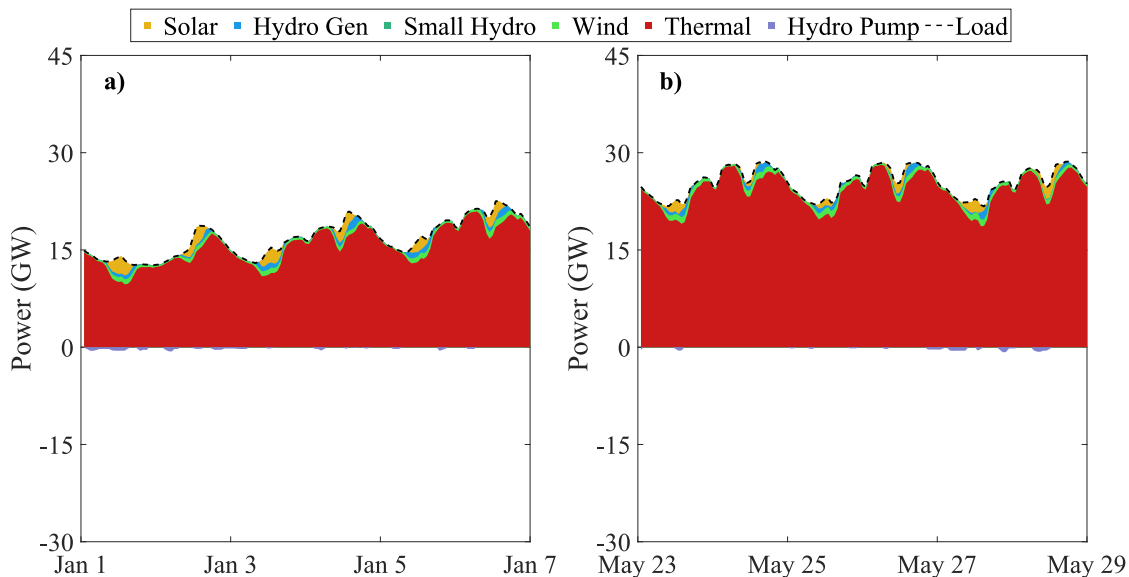


Fig. 5. Comparison of Actual EGAT's Hourly Power Generation Profiles in Scenario 1 during Load Demand Changes: (a) Lowest load demand period and (b) Highest load demand period.

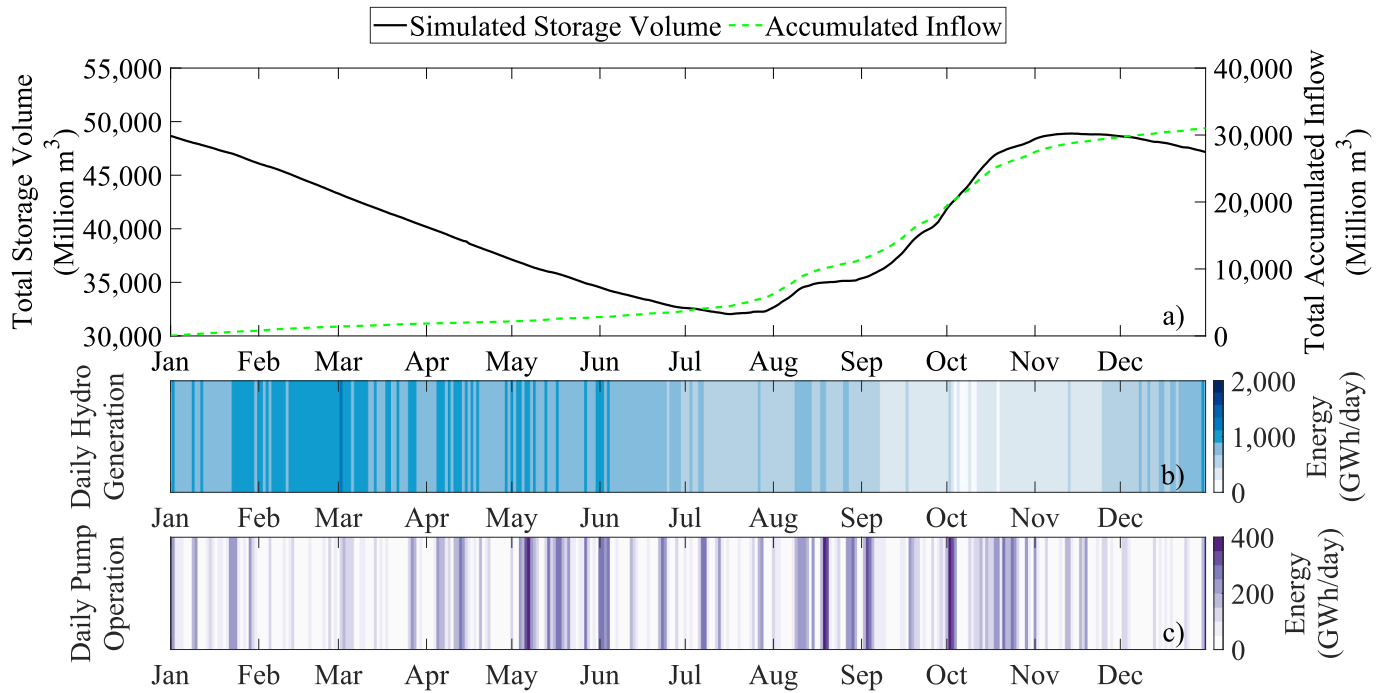


Fig. 6. Hydropower and PSH Operation throughout the year of 12 major EGAT's reservoirs under Actual EGAT's Operation Scenario: (a) Monthly Variation of Total Simulated Storage Volume and Total Accumulated Inflow, (b) Daily Total Hydropower Generation and (c) Daily Total Pump Operation.

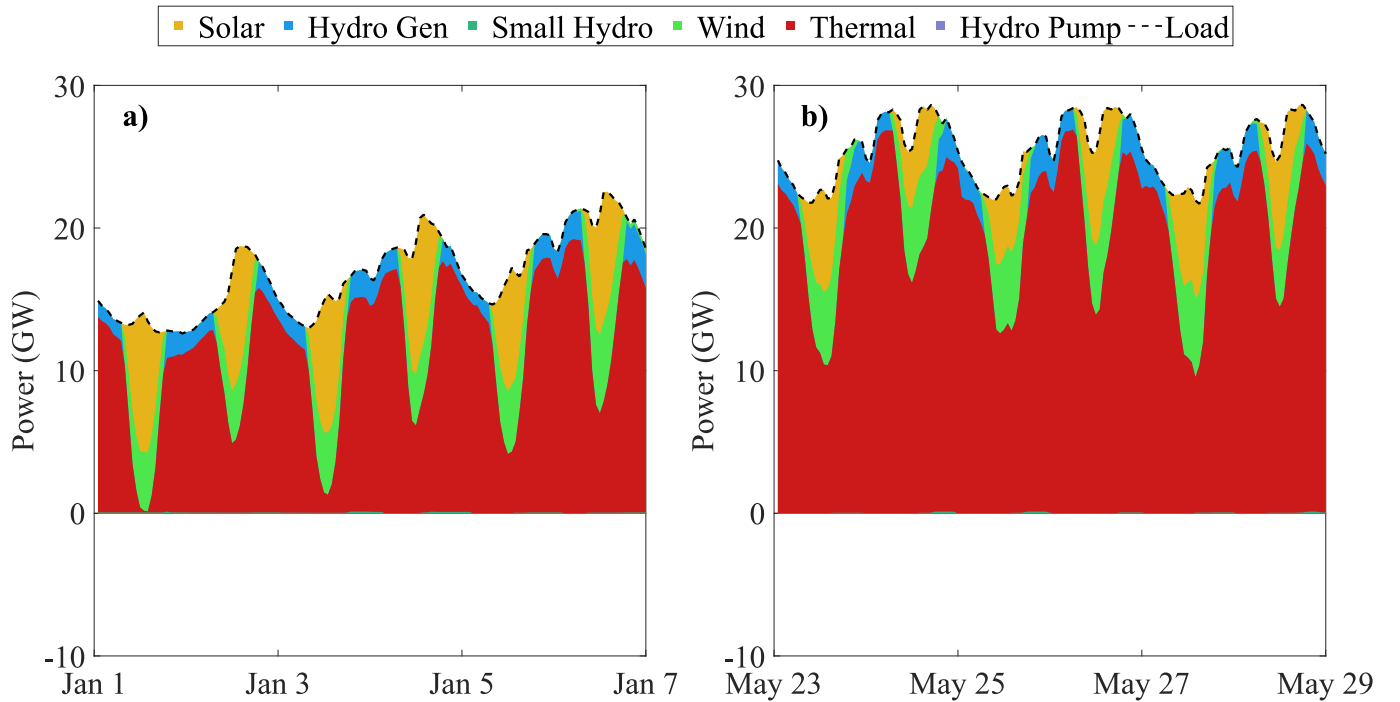


Fig. 7. Comparison of Optimized Operation without utilizing PSH Power Generation Profiles in Scenario 2 during Load Demand Changes: (a) Lowest load demand period and (b) Highest load demand period.

weakened—hydropower covered the residual load demand.

Fig. 10 reveals that, despite the differing operational strategies across the three scenarios, the seasonal changes of total storage volume remained broadly consistent with those in Fig. 6. In all cases, total storage volume began relatively high early in the year, declined progressively toward mid-year, and subsequently recovered during the latter months as inflows increased, indicating that reservoir operation

exerts only a limited influence on the overall seasonal storage pattern.

In addition, pumping operations demonstrated irregular and less frequent patterns throughout the year, with the highest pumping operation observed in October when storage volumes are at their lowest. Pumping mode operated by excess solar generation during midday was consistently scheduled 2–5 days before hydropower generation, which alternated with subsequent hydropower generation. Hydropower units

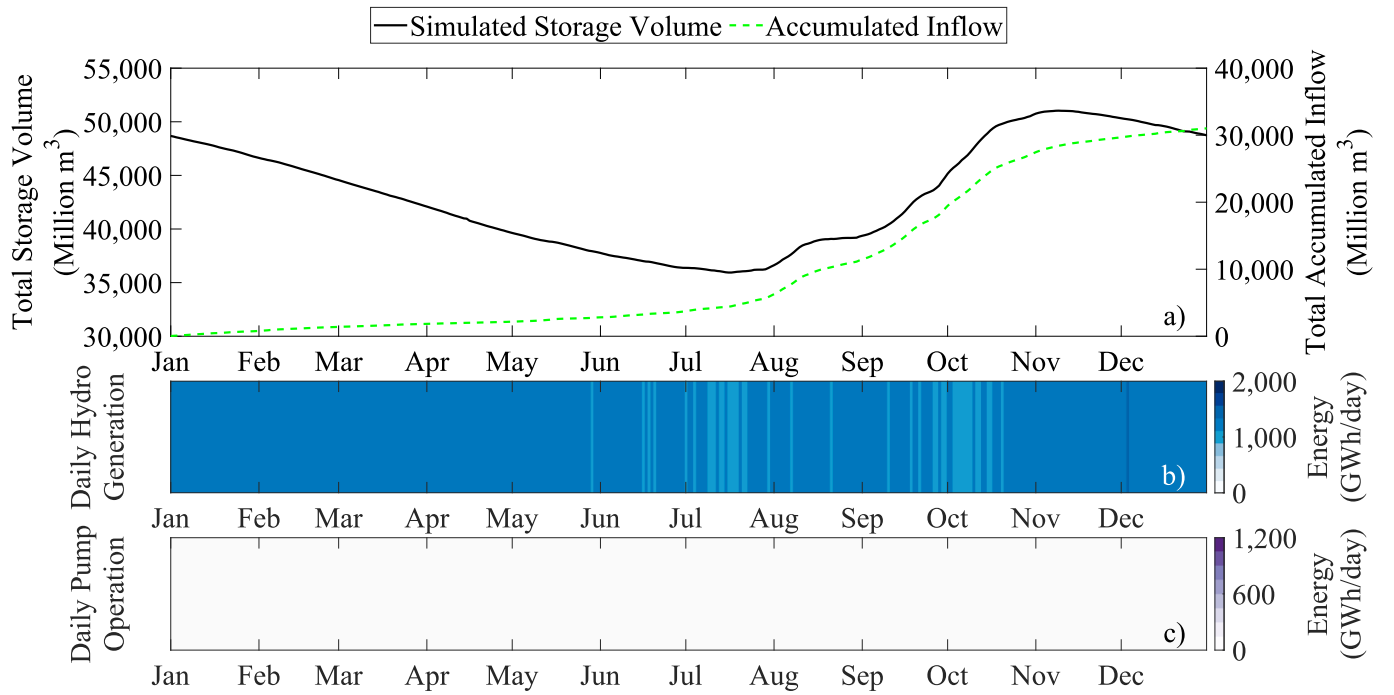


Fig. 8. Hydropower and PSH Operation throughout the year under Optimized Operation without PSH scenario: (a) Monthly Variation of Simulated Storage Volume and Accumulated Inflow, (b) Daily Hydropower Generation and (c) Daily Pump Operation.

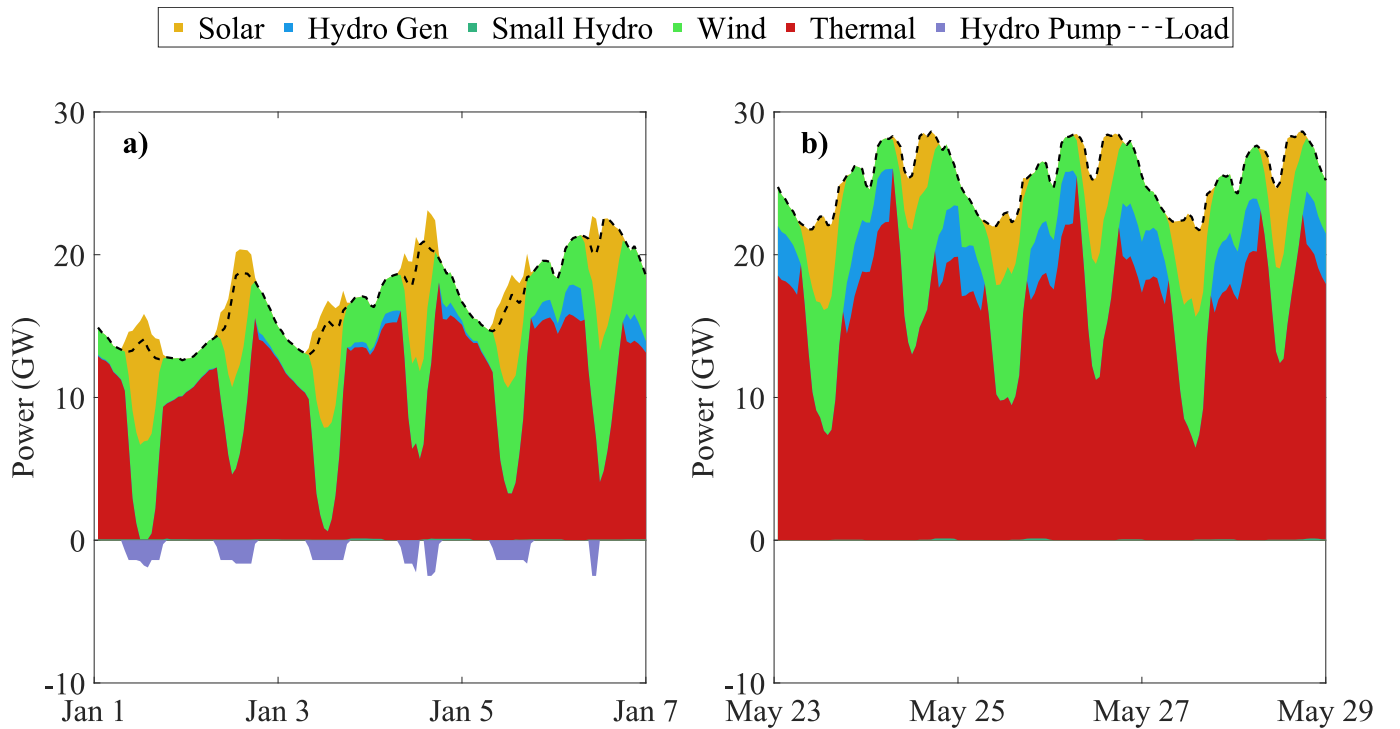


Fig. 9. Comparison of Optimized Operation with utilizing PSH Power Generation Profiles in Scenario 3 during Load Demand Changes: (a) Lowest load demand period and (b) Highest load demand period.

typically operated continuously for 9–12 days before transitioning to the next cycle. Accordingly, the findings indicate that seasonal dispatch is not a governing driver of the optimized hydropower–PSH scheduled operation in Thailand. By contrast, the Actual EGAT's operation is season-dependent, because irrigation and agricultural requirements constrain reservoir releases and, consequently, unit dispatch.

4.2. Optimum installed capacity of solar and wind for case studies

This section evaluates how different installed capacities of solar and wind power, under three distinct operational scenarios in Fig. 5 to Fig. 10, affect both the RE share and the overall LCOE of the energy system. The analysis compares the existing installed capacities in Scenario 1 with the optimal capacities in Scenario 2 and Scenario 3 derived

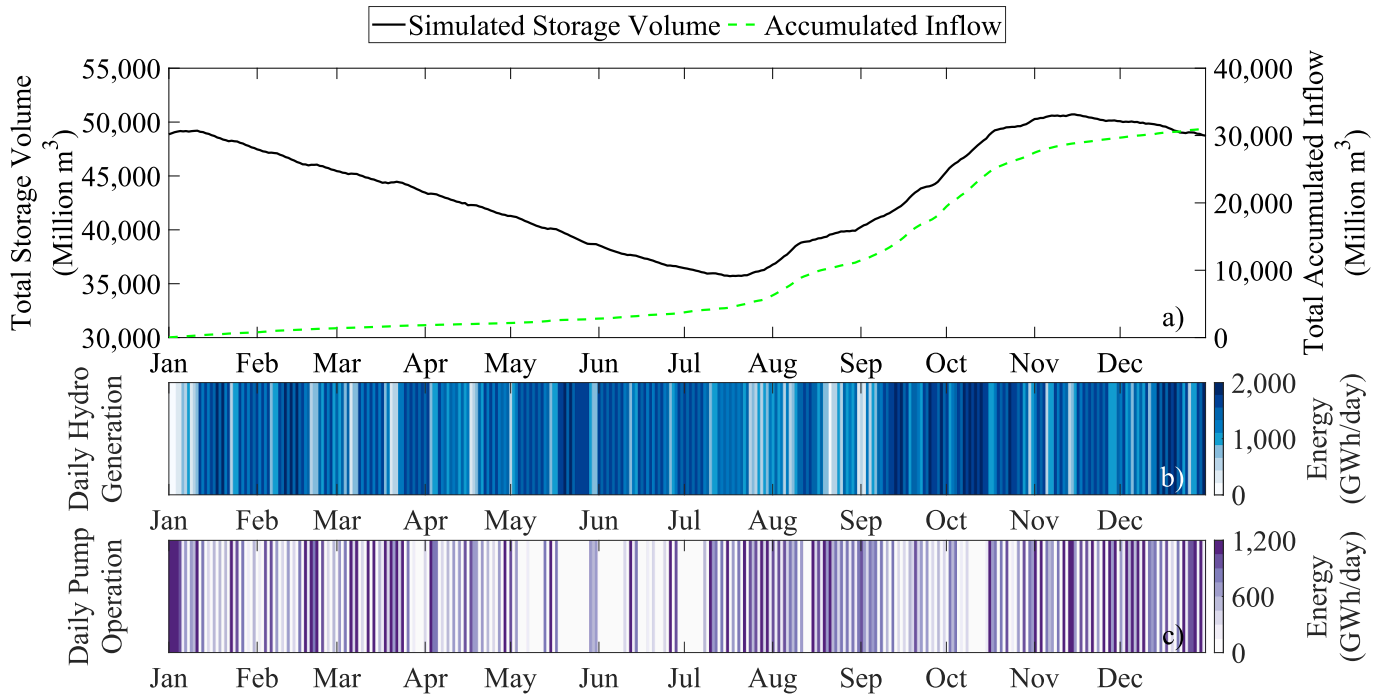


Fig. 10. Hydropower and PSH Operation throughout the year under Optimized Operation with PSH scenario: (a) Monthly Variation of Simulated Storage Volume and Accumulated Inflow, (b) Daily Hydropower Generation and (c) Daily Pump Operation.

from scenario-based optimization to examine the trade-offs between increasing VRE installed capacities and LCOE as shown in Fig. 11.

The optimization results in Fig. 11, consistent with the scenario design in Section 3, reveal a trade-off between RE share and LCOE as solar PV and wind installed capacities increase. In the **Actual EGAT's Operation scenario**, the installed capacities of solar PV and wind power were 3608 MW and 1774 MW, respectively. These values increased to

11463.23 MW and 9364.61 MW, respectively, under the **Optimized Operation without PSH scenario**, and further increased to 10423.31 MW and 15527.42 MW, respectively, in the **Optimized Operation with PSH scenario**. This progressive capacity expansion raises the RE share from 8.70% in the Actual EGAT's Operation scenario to 29.57% in the **Optimized Operation without PSH scenario**, with a corresponding moderate increase in LCOE from 1.37 to 1.68 THB/kWh. Further

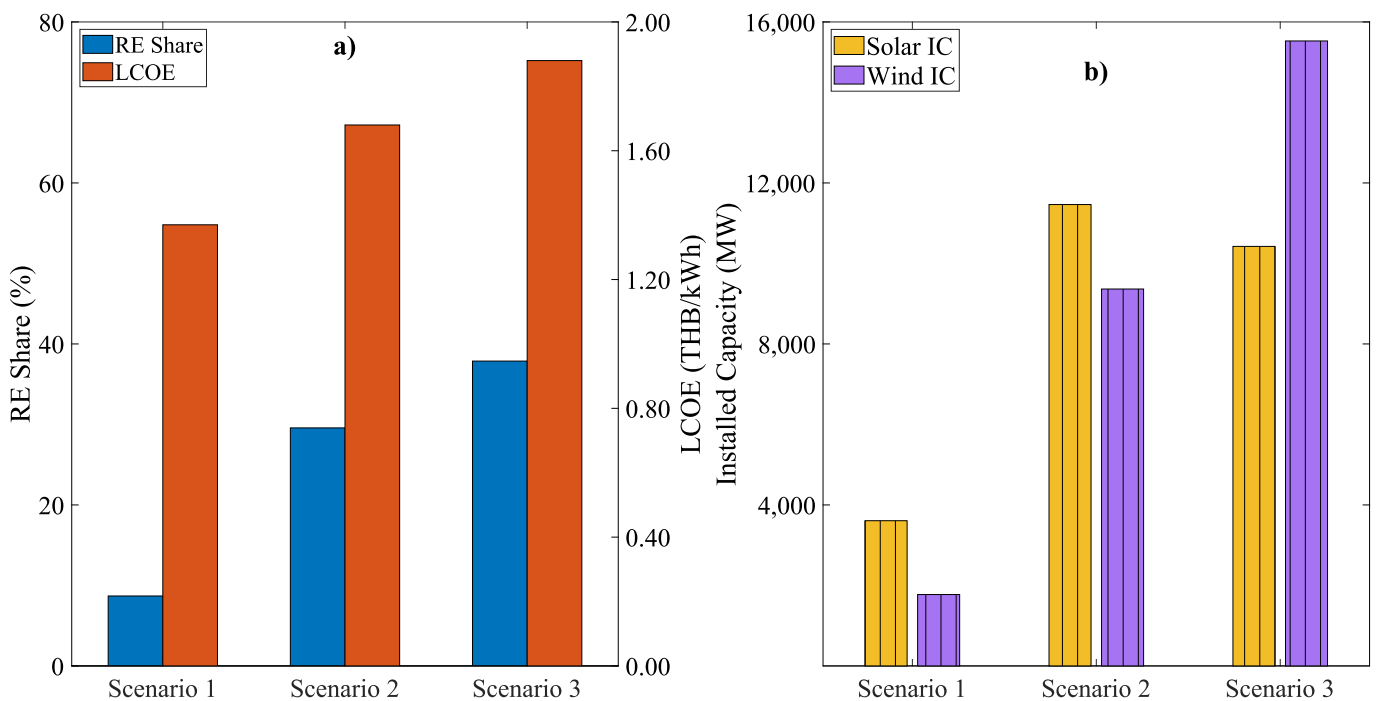


Fig. 11. Installed Capacity for Solar and Wind under Three Scenarios: Scenario 1 Actual EGAT's Scheduling Operation, Scenario2 Optimized Operation with Pumped Storage Hydropower, and Scenario 3 Optimized Operation with Pumped Storage Hydropower: (a) Comparison of RE Share and LCOE and (b) Comparison of Solar and Wind Installed Capacity.

expansion to a 37.87% RE share in the Optimized Operation with PSH scenario resulted in an LCOE of 1.89 THB/kWh. These findings indicate that higher RE shares, while achievable through increasing capacity expansion, are accompanied by diminishing economic returns, emphasizing the importance of balancing RE share targets with the overall LCOE of the system.

To the best of the authors' knowledge, no prior study has adopted a framework identical to that proposed in this work. The closest comparable previous works are: National-level studies such as Bissiri et al. [69], which assess RE share using installed VRE capacity as a percentage of peak demand and calculate curtailment mainly from transmission congestion, without evaluating LCOE. They typically reveal that much higher RE share is greater than 40% - 50% because they rely on VRE installed capacities with percentage of peak demand, which do not represent the actual renewable energy supply to the actual load demand. This differs fundamentally from the present study, which evaluates energy-based RE share from hourly power-balance dispatch, evaluates curtailment from excess VRE generation, and incorporates LCOE. Conversely, Regional single-dam studies such as Liu et al. [34], which analyze hydropower operation with PSH using power-balance formulations but do not aim to reduce curtailment to maximize RE share. Therefore, comparisons with existing studies should be interpreted in different contexts.

4.3. Sensitivity analysis of levelized cost of electricity

This section presents a sensitivity analysis of the Levelized Cost of Electricity (LCOE) to examine how variations in key techno-economic parameters influence the economic viability of the proposed renewable-hydropower-PSH system. The sensitivity analysis presented in this study assesses how economic parameters impact key techno-economic parameters on LCOE changes across the three scenarios as shown in Fig. 12. Six different economic parameters were varied by $\pm 10\%$, including initial investment cost (IC), fuel cost (FC), heat rate (HR), discount rate (r), total O&M cost (OM), and plant lifetime (L).

The charts reveal three insights. Firstly, IC shows the strongest impact on LCOE in all scenarios, particularly in Scenario 3, where a $\pm 10\%$ change results in the largest deviation. This is attributed to the high capital investment of the energy system, where capital expenditure was mainly impacted throughout lifetime. Secondly, FC and HR also show substantial influence, especially in Scenarios 1 and 2, presenting the residual dependence on thermal generation to supply unmet load. An increase in FC or HR proportionally increases fuel expenditure, leading to a noticeable increase in LCOE. Thirdly, discount rate and O&M cost show a moderate impact on LCOE. The discount rate has particularly less impact on Scenario 1. The total O&M has a relatively fewer impact throughout the lifetime compared to others, while plant lifetime showed the smallest changes. Increasing lifetime distributes the initial capital investment cost across more years, and increases total energy served load demand, resulting in lower LCOE. These emphasize the need to make educated decisions to reduce economic impacts on LCOE.

5. Limitation of this study

This study mainly focuses on the use of PSH to maximize the RE share and is subject to several limitations. First, the model does not account for competing water demand from the Royal Irrigation Department (RID), whose primary objective is agricultural water allocation. Second, the solar irradiance and wind speed data used in this analysis were obtained from nationwide meteorological stations across 60 provinces and 130 stations. The model does not explicitly incorporate site-specific resource characteristics or the spatial distribution of areas with high solar and wind potential. Although wind turbines generally yield higher annual energy generation than solar PV, the practical deployment of wind farms requires substantially larger land areas, which limits their feasibility in many regions of Thailand. Third, the operation of the pumping mode is simplified using a stepwise modular dispatch strategy. This modeling approach differs from actual EGAT PSH operations, where real-time electricity demand and water allocation requirements governed by the RID primarily determine dispatch decisions. In addition, Unit

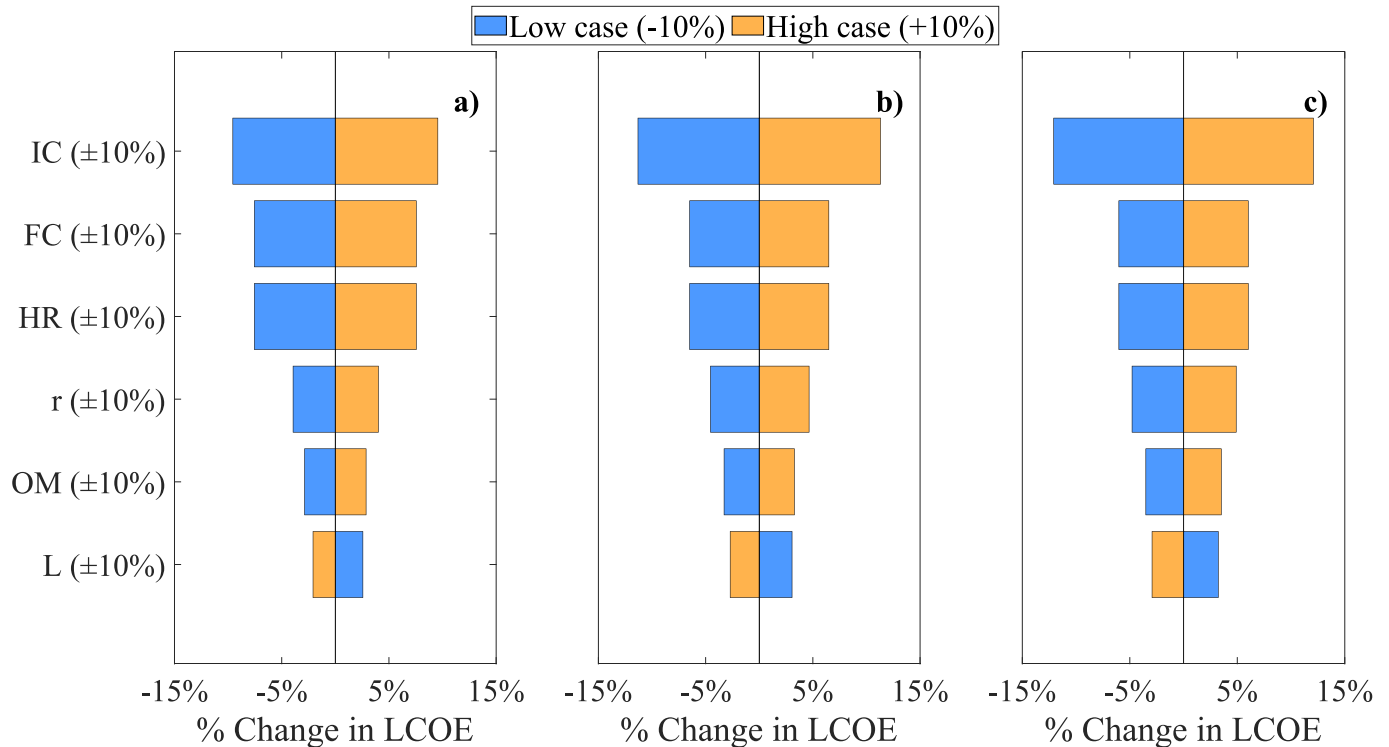


Fig. 12. Comparative sensitivity analysis of LCOE under different $\pm 10\%$ changes in initial investment cost (IC), Fuel Costs (FC), Heat Rate (HR), Discount Rate (r), Total O&M Costs (OM) and Lifetime (L) for different scenarios: a) Scenario 1, b) Scenario 2, and c) Scenario 3.

Commitment (UC) constraints are excluded because of the operational characteristics of PSH units in Thailand. According to the Contract Technical Specification for the Lamtakong Jolabha Vadhana Hydropower Plant (Units 3–4) PSH Project, the start–stop time of PSH units ranges from 2 to 5 min, which is considerably shorter than the one-hour resolution used in this study. Therefore, start-up/shutdown delays and minimum on/off durations have negligible influence on hourly hydropower and PSH dispatch under the considered framework. Given the fast response capability and instantaneous mode switching of PSH units, excluding these UC constraints is not expected to materially alter the scheduling results or the LCOE and RE share obtained in this study.

Furthermore, the LCOE estimation employed in this study does not adopt the tariff structure regulated by the Energy Regulatory Commission (ERC) of Thailand and excludes financial components such as financing charges, loan requirements, debt repayments, and other investment-related costs. Consequently, the LCOE results represent technical-economic performance rather than return on invested capital (ROIC). Nonetheless, the limitations associated with this assumption do not significantly affect the evaluation of system adequacy or electricity supply performance, which constitute the main focus of the study.

6. Conclusions and future work

This study develops and implements a novel national-scale optimization framework to enhance the coordinated scheduling of Thailand's solar, wind, hydropower, and PSH systems. By integrating a Hybrid Genetic Algorithm–Particle Swarm Optimization (GA–PSO) with CS dimensionality reduction, the model effectively addresses the high-dimensional, nonlinear nature of hourly dispatch optimization under real-world hydrological constraints.

Simulation results across three operational scenarios demonstrate that RE share can be increased from 8.70% under **Actual EGAT's Operation scenario**, which is heavily dependent on thermal power as the primary energy source, to 29.57% through **Optimized Operation without PSH scenario**. Further integration of PSH under **Optimized Operation with PSH scenario** raises the RE share to 37.87% by enabling energy shifting, reducing curtailment, and supporting reservoir management. However, increasing the installed capacities of solar PV and wind leads to a higher LCOE, rising from 1.37 THB/kWh in the base case to 1.89 THB/kWh in the high-RE PSH-integrated scenario. While the LCOE of individual technologies such as solar PV and wind remains low, system-wide LCOE increases due to the costs of overcapacity and storage infrastructure. These findings highlight a fundamental trade-off: pursuing higher RE shares improves sustainability but may incur diminishing economic returns. Hence, optimal RE deployment must carefully balance other objectives with cost-effectiveness. The model also highlights key operational challenges, such as early-year water release constraints, inflow-induced spillage risk, and thermal dependency during solar-wind deficits. These challenges are mitigated under the proposed optimized PSH-integrated scenario, which maintains storage volume balance, enhances RE utilization, and reduces LCOE.

Future research will extend this model by incorporating transmission constraints and market-based dispatch. Including stochastic programming techniques could improve handling of uncertainty. Moreover, evaluating additional storage (e.g., Battery Energy Storage System) would provide further flexibility. Integrating these elements will support robust long-term planning for Thailand's high-renewable future.

CRedit authorship contribution statement

Thanaphon Mathuravech: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Saran Salakij:** Writing – review & editing, Validation, Supervision, Software, Project administration, Methodology, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Google's Gemini and Paperpal in order to assist writing process to improve the readability and language of the manuscript. After using this tool/service, the authors reviewed and edited the content as needed and took full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors gratefully acknowledge the Electricity Generating Authority of Thailand (EGAT) and the Thai Meteorological Department for providing the essential datasets used in this study. We also extend their appreciation to all individuals who contributed to the discussion and review of the work.

Data availability

Data will be made available on request.

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